

IEEE

potentials

THE MAGAZINE FOR HIGH-TECH INNOVATORS

January/February 2025, Vol. 44 No. 1



Emerging Technologies and Dynamic Career Horizons

In this issue

- Quantum Communication Networks
- Digital Twins
- Social Robots
- Employment Dynamics



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ON THE COVER:

In this issue we investigate the forefront of breakthrough technologies set to revolutionize various sectors and we examine how these pioneering developments are creating novel career avenues and reshaping the employment landscape.

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Engineering the future

by Zoya Shafique^{ID}

Charles Dickens said it best: “It was the best of times, it was the worst of times.” This sentiment holds just as true today. We live in an era of rapid technological advancement: breakthroughs in artificial intelligence (AI), renewable energy, and semiconductor design are shaping the future at an unprecedented pace. Yet, we also face significant global challenges: climate change is intensifying, misinformation is spreading rampantly, and geopolitical conflicts are destabilizing entire regions and countries.

Looking around, one cannot help but feel overwhelmed. With so many events unfolding across the globe, it is tempting to wonder, “What can I do?” But if everyone falls into the trap of believing that one person alone cannot make a difference, then no one would take action at all. Instead, this is exactly the time to engage, to educate ourselves, to innovate, and to build stronger, more resilient communities.

Technology is at the heart of both the problems we face and the solutions we can create. As engineers, we have the ability, and the responsibility, to shape the future, regardless of scale, because even small contributions matter. Direct impact in your community is just as important as large-scale change. Maybe your research

on low-power electronics leads to more sustainable devices. Maybe your work in embedded systems helps improve disaster response technology. Maybe your efforts in promoting ethical AI development ensure that automation benefits society rather than deepens inequality.

It is also our duty to create dialogue within our communities on pressing issues and to think critically on the benefits and harms of the technologies we are working on. For instance, some AI proponents claim that AI will help solve climate change. To them, I ask, what about all the destruction to the environ-

ment being caused in the process of creating an AI to solve climate change? Our ability to question, analyze, and innovate responsibly is what makes engineers vital to shaping technology’s impact on society.

It is easy to dismiss individual efforts, but this is how we lose. Instead, we must stand firm, commit to ethical engineering practices, and use our knowledge to drive positive change. As future engineers, we are not just problem solvers; we are the architects of tomorrow. Let us make sure it is one worth building for all citizens of the world.

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Quantum communication networks

Design, reliability, and security

Brian Hildebrand, Ashutosh Ghimire, Fathi Amsaad^{ID}, Abdul Razaque, and Saraju P. Mohanty^{ID}

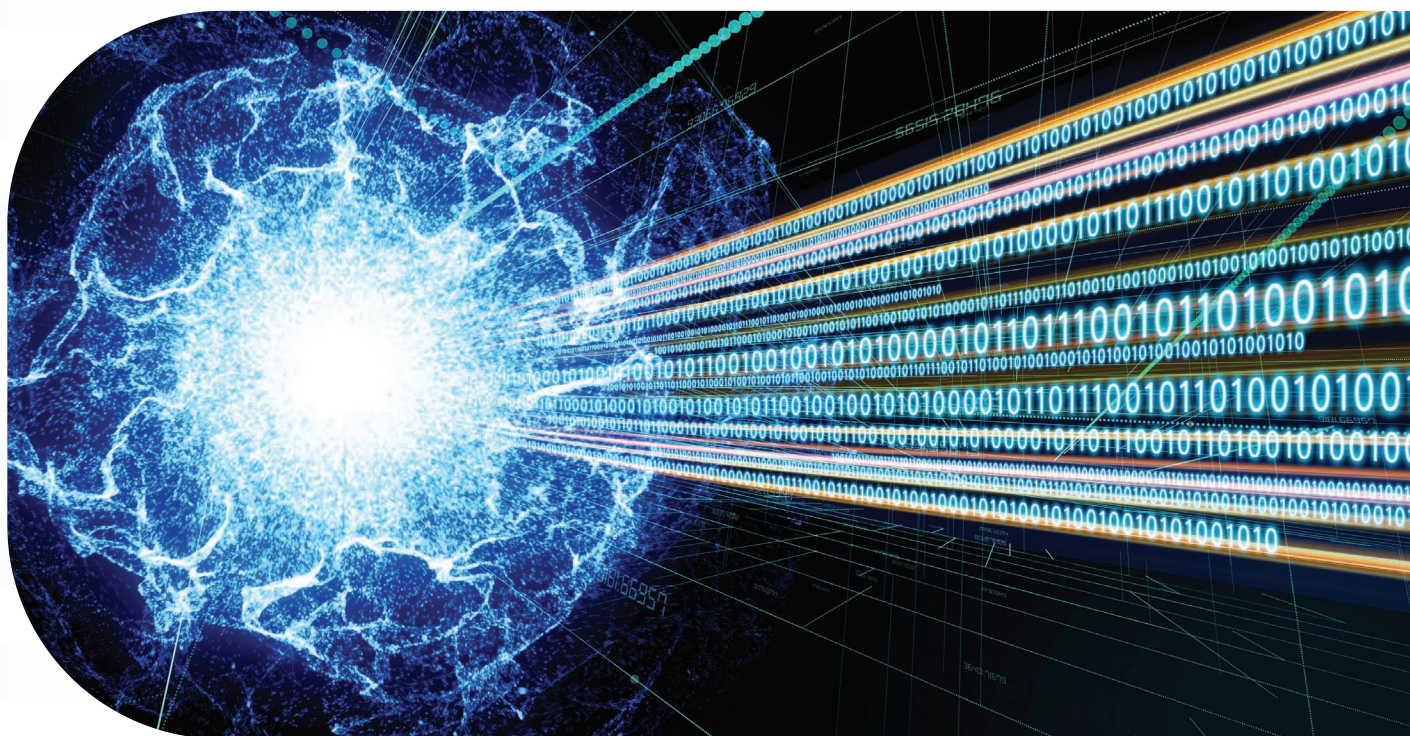
The overall purpose of this study is to explore the potential of quantum-based communication networks, leveraging the unique properties of quantum entanglement and superposition, to address critical applications, such as image processing, machine learning, data mining, and encryption. Additionally, this research aims to overcome the limitations of traditional binary

systems in the field of computing. The basic methodology of this research involves a comprehensive investigation into the establishment of end-to-end entanglement between two quantum computers, which is fundamental for building robust quantum networks. Performance measures for quantum computers within the context of quantum networks are defined and evaluated, with a specific focus on network design, reliability, security, and trust. Quantum-dot cellular automata (QCA) devices are introduced as a technological advancement to

enhance security and trust mechanisms within quantum communication networks.

Furthermore, the study addresses network congestion issues and outlines quantum-based solutions while also examining emerging standards for their assessment. A distinctive contribution of this research lies in the exploration of quantum key distribution (QKD) and the utilization of advanced cryptosystem methodologies to strengthen security and trust within quantum-based networks. In conclusion, this research seeks to bridge existing gaps by addressing

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the challenge of achieving end-to-end entanglement, evaluating critical performance metrics, introducing innovative technologies, and proposing enhancements to security and trust mechanisms. The findings of this study hold promise for not only advancing the field of quantum networking but also impacting a wider range of applications that require secure and future-oriented communication solutions in an increasingly interconnected world.

Introduction

The next-generation technology has paved the way for the realization of quantum networking, enabling efficient communication between quantum devices to solve complex problems more effectively than traditional computing algorithms. In the realm of next-generation quantum networking, researchers have explored harnessing quantum computer features, such as circuit design, entanglement, and superposition, to accelerate specific calculations.

This article begins by defining quantum computing and its inherent features, such as quantum design, entanglement, and superposition, among other attributes that enhance specific communications. We delve into the concept of quantum entanglement, a distinctive trait of quantum computing involving quantum bits (qubits) (Fig. 1) that exhibit strong correlation. Furthermore, the article demonstrates how measuring a qubit leads to its collapse into one of two potential values, similar to a classical binary digit. Our discussion extends to quantum routing (Fig. 2), highlighting the coordination of entangled qubits and their correlated movement. Ultimately, the article provides a comprehensive view of performance measures for evaluating quantum networks, encompassing quantum routing, scheduling, reliability, security, and standards for QKD protocols.

Figure 1 illustrates a generalized quantum network comprising both quantum and classical communication channels. This showcases a tele-

portation technique that establishes end-to-end entanglement between quantum computers, a pivotal operation for constructing basic quantum networks. Multiple quantum repeaters integrated along quantum channels extend the entanglement range, enabling connectivity over greater distances. As seen in Fig. 1, teleportation is crucial in quantum networking, allowing quantum circuits to extend across computers. This entails entangling qubits across the network, interconnecting qubits between computers. These interlinked qubits, referred to as *flying qubits*, encode photon spin or polarization, traversing optical fibers or wireless mediums. Thoughtfully sequenced flying qubits facilitate quantum state transfer across extended distances.

At the core of a quantum computer lies the qubit, analogous to the classical binary digit (bit). Unlike classical bits that represent only zero or one, qubits span the full range between zero and one, justified by the principles of quantum mechanics. In classical computing, a bit can ex-

ist in one of two discrete states, zero or one. However, in quantum computing, qubits can exist in a state of superposition, meaning they can represent a combination of both zero and one simultaneously as well as any value between these two states.

The concept that qubits can exist in a superposition of states between zero and one is a fundamental aspect of quantum mechanics and serves as a foundation for quantum computation. This idea is often explained through Schrödinger's equation, which describes how quantum states evolve over time. In quantum computing, qubits leverage this property to perform multiple calculations simultaneously, leading to the potential for exponential speedup in certain computations compared to classical computers.

The subsequent sections are organized as follows. The next section presents an overview of quantum design, followed by sections covering quantum network routing, quantum network reliability, and quantum communication security. The final section concludes the article.

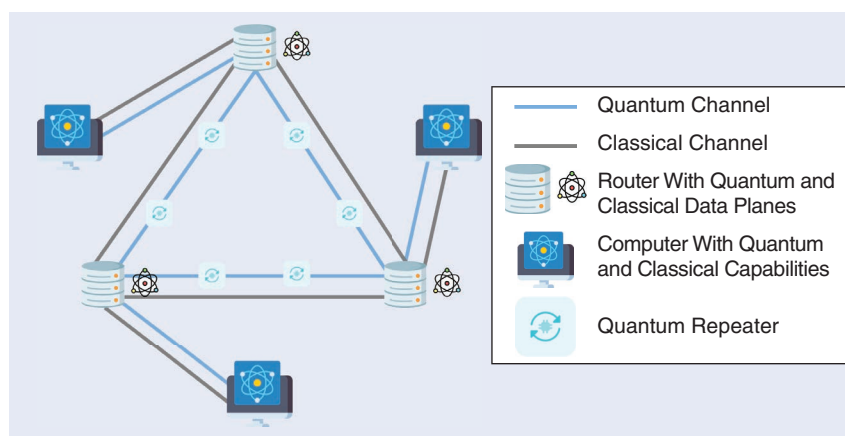


FIG1 A quantum network with quantum and classical channels. Quantum repeaters are used to extend the range of quantum entanglement.

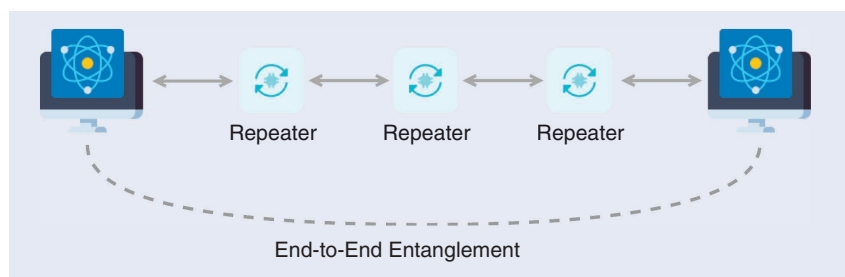


FIG2 The end-to-end entanglement between two quantum computers using a series of three quantum repeaters.

Quantum design

Implementing quantum algorithms within quantum-based computers requires designing quantum circuits for executing arithmetic operations, including addition, subtraction, and multiplication. Our journey into quantum algorithms, covering tasks such as integer factoring, searching, and quantum mechanical modeling, begins with the quantum circuit for multiplication. Qubits serve as the fundamental unit of quantum infor-

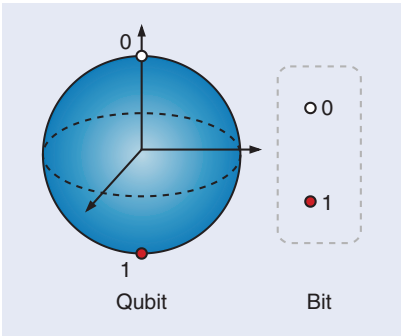


FIG3 A graphical representation of a qubit and a traditional binary digit (bit).

mation. In Fig. 3, we illustrate the distinction between the conventional binary digit (bit) and the qubit. While a bit can only adopt values of zero or one, a qubit can represent zero or one as well as any value between them or on the surface of a sphere. A distinctive attribute of the qubit is its superposition property, wherein it exists in multiple states until measured. Following measurement, the qubit collapses to a specific state. Given the limited availability of qubits in existing quantum computers, researchers often consider the qubit cost, which represents the total number of qubits used in constructing a quantum circuit, as a critical performance measure.

However, quantum computers face a significant challenge in increased susceptibility to noise errors, known as *decoherence*. This phenomenon deteriorates qubit state functions, reducing the accuracy of quantum calculations. Decoherence arises from entanglement with external

factors, such as light, vibration, sound, cosmic rays, and interference from wireless devices, as well as internal interactions between qubits within the same system. The task of isolating individual qubits further compounds this challenge. A potential solution involves constructing quantum circuits that incorporate Clifford + T gates, leveraging their inherent fault tolerance capabilities. Clifford gates, which use conjugation to normalize qubits, offer a means of error mitigation. When combined with T gates, these circuits gain resilience through the application of error-correcting codes, a vital technique for safeguarding quantum information from noise-induced errors.

The introduction of fault tolerance brings with it additional challenges of computational overhead. Unlike classical bits with discrete “1” or “0” states, qubits exist in superposition. Maintaining fault tolerance necessitates resources to restore qubit states, which extends to the complexity of T gate implementation. Managing this supplementary layer of overhead is crucial for ensuring the reliability and robustness of quantum computations.

Table 1 provides a summarized comparison among different quantum networking performance measures. QCA offers an energy-efficient solution for circuits with limited resources. QCA utilizes quantum circuits and internal connections to interlink building blocks, enabling the design of low-power circuits and high-throughput networks. In Fig. 4, you can see an overview of a typical QCA configuration, with shaded dots representing two charged electrons capable of tunneling across the cell. Governed by electrostatic repulsion, these electrons strive to position themselves as far apart as possible within the cell, representing two distinct states. Recent research highlights QCA’s potential to replace conventional integrated circuits, making it a compelling option for low-power circuit design.

In the realm of digital circuits, an adder is responsible for performing addition operations. Among these,

TABLE 1. A comparison among different quantum networking performance measures.		
QUANTUM NETWORKING RESEARCH CHALLENGE	QUANTUM PERFORMANCE MEASUREMENT	PROPOSED QUANTUM APPROACH AND IMPLEMENTATION
Quantum circuit design	T count	T gates employed to build quantum design
	T depth	Layers of T gates in quantum design
	Qubit cost	Number of qubits used to build a quantum circuit
Quantum routing	Entanglement fidelity	Measure of the difference between the actual and desired states of a quantum system
	Throughput	Rate of successful delivery of data across a network
Quantum solutions to network congestion	Routing time	Time it takes a packet to choose a route
	Travel time	Time it takes a packet to travel to the destination
	Total transmission time	Sum of routing time and travel time
QKD for 5G	Fault tolerance to noise	Effect of noise on qubit states
	Latency	Communication delay over a network
	Bandwidth	Maximum amount of data transmitted over a network in a given timeframe
	Massive connectivity	Ability to connect to a large amount of devices/computers

the carry look-ahead adder (CLA) stands out as a pivotal component in digital arithmetic processing. The CLA excels in its ability to perform addition with higher efficiency and speed compared to alternatives, such as the ripple carry adder (RCA). This efficiency stems from its capability to predict carry bits before determining the sum, effectively reducing computation time.

The quantum CLA (QCLA) serves as the quantum counterpart to the CLA, executing addition with a time complexity of $O(\log(n))$. The QCLA leverages RCAs and is characterized by a linear depth equivalent to its input bit count. This results in significantly improved speed compared to RCAs, which require $O(n)$ time for similar operations. QCA offers an energy-efficient solution for circuits with limited resources. By utilizing quantum circuits and internal connections, QCA can interlink building blocks, enabling the design of low-power circuits and high-throughput networks. The general QCA setup, as depicted in Fig. 4, features charged electrons capable of tunneling across the cell. Governed by electrostatic repulsion, these electrons assume positions that maximize their separation within the cell, representing distinct states.

Direct connections between inputs and outputs of a quantum circuit often introduce complexities,

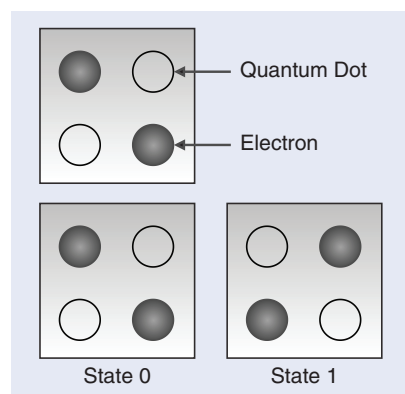


FIG4 QCA. The shaded dots represent two charged electrons free to move to any part of the cell. Based on the principle of electrostatic repulsion, the charged electrons will attempt to move as far as possible from each other in the cell. The two electrons represent two possible states.

necessitating ancillae (auxiliary qubits) and producing output garbage. Ancillae are involved when a constant input is applied to the quantum circuit, while garbage output arises when a quantum circuit's work does not contribute as input to another circuit. Removing garbage output is crucial for reducing the quantum circuit's overall qubit and T gate costs. QCLAs can be categorized as in place and out of place. An in-place QCLA replaces a direct input with the sum, while an out-of-place QCLA returns the sum on ancillae. For quantum computing applications, QCLAs should, ideally, produce no garbage output without significantly impacting qubit and T gate costs. Achieving this goal requires efficient utilization of T-gate-efficient Toffoli gates, which serve as reversible logic building blocks.

Quantum network routing

Efficiently constructing quantum-based network topologies necessitates the creation and optimization of network routing. Quantum routing is centered around identifying the optimal channel for teleporting qubits, a process crucial for extending quantum circuits beyond individual computers. Unlike classical bits, qubits cannot be copied, but they can be teleported between quantum systems, establishing quantum entanglement. This unique feature, characterized by strong correlations among specific groups of qubits, demands a holistic consideration of potential qubit values due to interdependent probabilities.

Entangling qubits is facilitated by "flying qubits," which encode states in photon spin or polarization for transmission through fiber-optic cables or wireless methods. However, flying qubits experience fidelity degradation with increased distance. Quantum repeaters serve as intermediaries to extend their range, utilizing entanglement swapping between two flying qubits to generate

end-to-end entanglement. Challenges, such as signal power attenuation and transmission imperfections, influence entanglement quality along the chain. Evaluation measures for quantum network quality encompass throughput and entanglement fidelity, reflecting data delivery success rates and the variance between desired and actual quantum states.

In Fig. 2, you can see the illustration of the establishment of end-to-end entanglement between quantum computers via three quantum repeaters. This emphasizes the significance of quantum entanglement and repeaters within networks that integrate both quantum and classical communication. The efficiency of quantum entanglement relies on the effectiveness of quantum routing strategies, which involves finding the right balance between scheduling and path selection. Scheduling entails selecting quantum couples to establish end-to-end entanglement within specific time slots. The choice of optimal paths or quantum repeater sets plays a vital role in entanglement creation. Effective quantum routing utilizes strategies such as weighted round-robin or iterative selection for path optimization, enhancing both the entanglement rate and fidelity.

Certain quantum routing algorithms preallocate qubits to prevent resource conflicts during entanglement swapping. However, resource fragmentation can hinder network efficiency. Strategies like Quantum CAST Algorithm (Q-CAST) aim to mitigate this issue by considering fragmentation during entanglement and using priority connections for recovery in the case of failures. Another approach involves employing entanglement purification techniques to enhance entanglement quality between sets of qubits across nodes. Quantum PATH Algorithm (Q-PATH) identifies purification choices based on cost considerations, while Quantum LEAP Algorithm

(Q-LEAP) minimizes fidelity loss along the shortest route, using efficient purification decision methods to streamline computations.

Quantum network reliability

In the realm of network communications, users and computers often compete for the shortest and quickest paths. However, network reliability becomes a concern when multiple users or computers choose the same route. The criteria for evaluating quantum solutions to network reliability are succinctly outlined in Table 1.

Navigating network reliability requires a delicate balance between transmission latency and cost. For example, network congestion can be assessed through the total transmission time, which combines the routing time and travel time. The travel time represents the duration for a network packet to reach its destination, while the routing time signifies the time taken for a packet to select a route. As more packets opt for a common route, the travel time for each packet increases. Therefore, each packet faces two choices: 1) opt for the shortest path and risk potential delays due to congestion, or 2) choose a longer path with lower congestion-induced delays.

Classical approaches treat these options as binary decisions (zero or one). In contrast, quantum computing allows for representing various options through multiple states between zero and one within a qubit. Additionally, the impact of noise on qubit states, termed *fault tolerance to noise*, is a crucial consideration. Novel proposed solutions leverage quantum game theory to address network congestion and reliability concerns. In this paradigm, packets engage in self-interested competition for optimal routes. Initially, quantum states are assigned to represent each possible strategy. Subsequently, each packet adapts its local qubit's state to choose the optimal strategy. Such approaches hold potential to reduce travel and routing times, providing innovative solutions to alleviate network congestion and enhance reliability.

Quantum communication security

Security concerns loom over end-to-end quantum networks, encompassing challenges in cybersecurity across cloud and edge computing, privacy and security attacks in fog computing architectures, and vulnerabilities in Internet of Things (IoT)-based devices. To address

these concerns, QKD emerges as a novel security paradigm, harnessing the unique properties of quantum mechanics for cryptographic operations, such as encryption and decryption. QKD also provides a means to obtain critical cryptanalysis information, serving as a countermeasure against attempts to exploit quantum-based physical characteristics for constructing distributed symmetric keys.

In Fig. 5, you can see a typical QKD scenario, where two computers establish a network link via classical and quantum channels. In this illustration, one computer generates a secret key and transmits it to the other computer through the quantum channel, established via end-to-end entanglement and teleportation techniques. The quantum nature of the channel ensures that any eavesdropping attempt results in the qubit's consumption upon measurement. Upon receiving an intact message containing the secret key, the second computer uses the private and public keys for decryption. If the integrity of the secret key is compromised, the key must be re-sent, ensuring robust security in the quantum communication network.

Table 1 provides a summary of evaluation criteria for QKD implementations in 5G networks. The key performance indicators of 5G networks include latency, bandwidth, and massive connectivity. Latency is the measure of communication delay over a network. Bandwidth is the maximum amount of data a network is capable of having transmitted over it at any given timeframe. Massive connectivity is the measure of a network's ability to connect to a large number of computers or devices. To meet the ambitious performance goals of 5G networks, including low latency, ample bandwidth, and widespread connectivity, researchers have embraced advancements like software-defined networks and network function virtualization (NFV) to rapidly deploy network services. While NFV orchestrators facilitate the creation and implementation

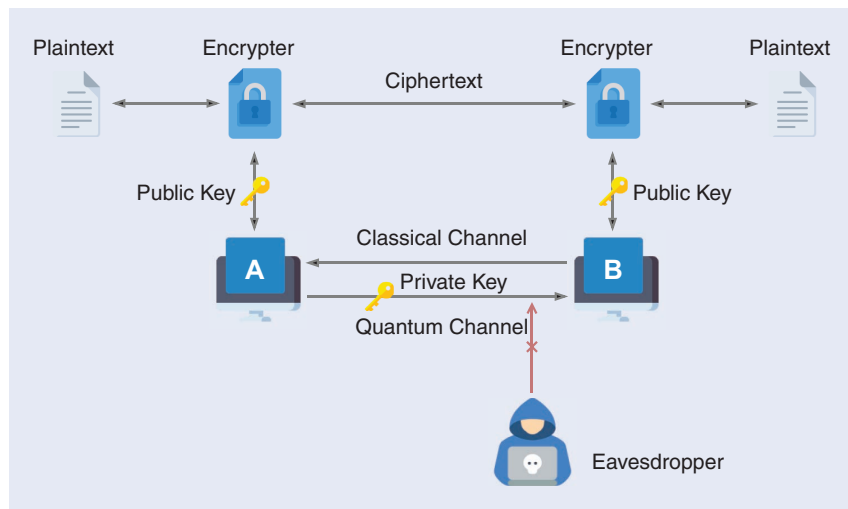


FIG5 QKD. Two computers, A and B, establish a connection via classical and quantum channels. A transmits the secret key to B over a quantum channel established using quantum entanglement and teleportation techniques. An eavesdropper cannot measure the qubits on a quantum channel without disturbing the integrity of the qubits. The loss of integrity of the message sent over the quantum channel is indicative of an eavesdropping attempt. B can use the private and public keys to decrypt if message integrity is good. Otherwise, the secret key needs to be present.

of services like NFVs, their current scope is confined to single administrative domains.

Progress is being made in the realm of multidomain orchestration, empowering service providers to establish comprehensive network services that span distinct network administration domains. However, this advancement introduces overarching security concerns that encompass the entire end-to-end network. These concerns include potential security threats associated with mobile edge computing and IoT devices as well as security and privacy vulnerabilities tied to fog computing architectures. Addressing these security challenges, some proposals advocate the use of QKD.

For instance, q-ROADM, a QKD switch-enabled flex-grid architecture, supports classical and quantum data channels across a 5G optical network. q-ROADM aims to enhance end-to-end security across heterogeneous networks, a significant objective of 5G. Expanding on the 5GUK Exchange orchestrator, the q-ROADM framework demonstrates a quantum-secured diverse multidomain 5G network. This network facilitates the seamless deployment of services by chaining virtual network functions (VNFs) across various domains. The architecture integrates a quantum-secured optical network, catering to both quantum and classical channels, and offers dynamic quantum key switching and inter-domain optical network connectivity. This approach enables efficient VNF chaining while ensuring robust quantum protection.

To bolster security and trust in future quantum and postquantum cryptographic standardization, cryptosystem approaches have emerged. These implementations, including Quantum Low Density Parity-Check (LDPC) and Quantum Moderate Density Parity-Check (MDPC), Reed-Solomon designs, and McEliece, address the security challenges of convolutional cryptography and safeguard quantum systems against cyberattacks. These techniques find applications in consum-

er electronics, artificial intelligence, and blockchain-based technologies like Bitcoin. The approaches involve generating binary codes for system authentication, secure digital signatures, and cryptographic key exchange over untrusted mediums, offering resilience against various attacks. Emerging cyberattacks targeting quantum computing, such as side-channel and power analysis attacks, are also being countered. One unique approach focuses on mitigating side-channel attacks during the implementation of the McEliece cryptosystem on resource-constrained platforms. Utilizing quantum arithmetic benchmarks, this approach evaluates cryptosystem measures benchmarked on field-programmable gate arrays, incorporating error detection mechanisms to thwart fault attacks in quantum computing design.

Conclusion

Quantum communication networks have emerged to allow devices to communicate efficiently, reliably, and securely as well as to perform critical tasks, such as image processing, science computations, and searching and breaking encryption. This work stands out due to the merit of its comprehensive exploration of quantum communication networks, offering a novel perspective on quantum computing's performance and security aspects. By delving into both foundational and advanced concepts, the article provides a unique blend of fundamental principles and cutting-edge techniques, showcasing its novelty.

The focus on quantum network security and design introduces a novel angle, addressing critical concerns in an evolving field. Additionally, the study's systematic approach, encompassing quantum circuit analysis, routing strategies, congestion solutions, and QKD integration, showcases the article's pioneering approach to addressing various challenges within quantum communication networks. The careful consideration of evaluation criteria throughout the study

adds a distinct merit, ensuring a rigorous analysis and presenting a practical framework for assessing quantum-based solutions. This article, therefore, offers valuable insights and contributions to advancing the field of quantum communication networks, both in terms of its merit and the innovative perspectives it presents.

Future work

Building upon the comprehensive exploration of quantum communication networks presented in this article, several avenues for future research and development emerge. First, further investigations can be conducted to enhance the practical implementation of the proposed quantum circuit designs, entanglement-based routing strategies, and congestion solutions. As quantum communication networks advance, the development of optimized protocols and algorithms could substantially improve their efficiency and reliability.

Moreover, the integration of emerging technologies, such as QCA, could be explored to enhance the performance and scalability of quantum communication systems. Research efforts could focus on developing QCA-based components that offer faster processing and improved error correction mechanisms, addressing the issue of decoherence and enhancing the overall security of quantum communication. Additionally, the integration of quantum communication networks with existing classical communication infrastructures presents a promising area for further investigation. Developing seamless interoperability protocols, efficient gateways, and hybrid network architectures could pave the way for practical implementations that leverage the strengths of both classical and quantum communication.

Finally, the study of quantum communication networks could expand to consider the implications of multiparty communication scenarios, enabling the development of robust and scalable protocols for secure multiparty quantum

communication. Further exploration of QKD methods, including advancements in postprocessing techniques, could lead to enhanced security protocols that withstand potential attacks and vulnerabilities.

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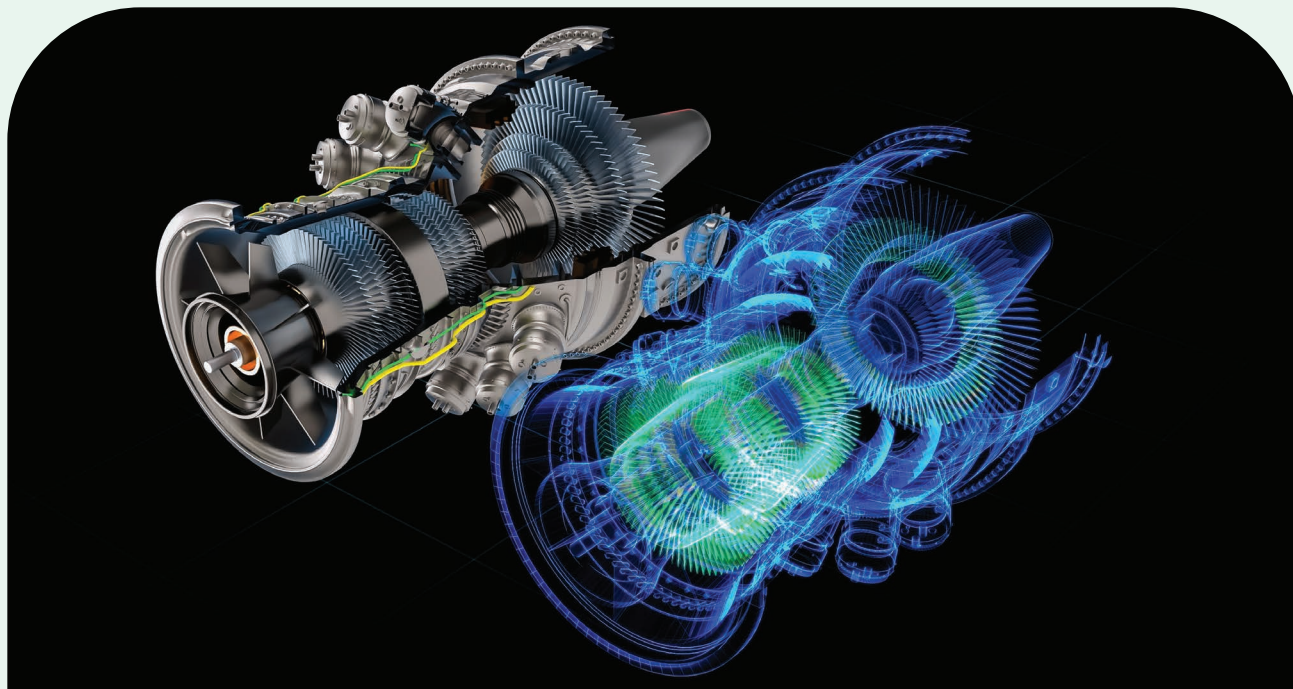
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Challenges in simulation of digital twins

A review

Dhanush L. Prakash, Nidhi B. C. , and Gopalakrishna H. D.



The digitalization of manufacturing processes is on the rise, marking the development of Industry 4.0 and the emergence of smart factories. Digital twins (DTs), which are digital replicas of physical assets or processes, are being utilized to enhance business applications. In the manufacturing sector, DTs can be created for asset-specific production lines or even entire production processes. Implementing the simulation of a DT in a manufacturing unit comes with its own set of challenges. It is crucial to integrate accurate and up-to-date

data from various sources, such as sensors and machines. Building an accurate model that accurately represents the behavior of the production line requires a deep understanding of its dynamics, interactions, and dependencies. Validating the model using real-world data is essential for obtaining reliable simulation results. This article aims to provide a review on DT technology, highlight the challenges faced during simulation, identify areas for improvement, and suggest future research topics in the field of DTs for manufacturing.

Introduction

A DT is a virtual representation of a physical system or process that

allows for real-time monitoring, analysis, and optimization. It serves as a digital replica that mimics the behavior and characteristics of its physical counterpart. In the context of smart manufacturing, DTs are being increasingly utilized to simulate and optimize production processes, predict and prevent equipment failures, and improve overall efficiency and quality.

The adoption of DTs in manufacturing is driven by their ability to provide a holistic, data-driven approach to process optimization, decision making, and increased efficiency. By creating a DT of a production line or system, manufacturers can gain a comprehensive understanding of its operations and identify

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areas for improvement. This enables them to make informed decisions and implement strategies that enhance productivity and reduce costs.

It is expected that the holistic use of DT models in product development and production will dominate future product generations because they allow the creation of high-performance products competitively (Botín-Sanabria et al., 2022). The global market of DT is expected to grow at a rate of 58%; by 2026, it is expected to reach US\$48.2 billion (Wärmefjord et al., 2020).

One of the key advantages of DTs is their predictive analysis capabilities. By continuously collecting and analyzing real-time data from sensors, machines, and other sources, manufacturers can anticipate potential issues and optimize processes accordingly. This proactive approach allows them to prevent downtime, optimize maintenance schedules, and ensure continuous production. Reducing downtime is crucial for manufacturers as it directly impacts productivity and profitability (Juarez et al., 2021). DTs play a vital role in this aspect by enabling proactive maintenance and predicting equipment failures. By analyzing sensor data and utilizing advanced analytics, manufacturers can identify potential issues before they occur and schedule maintenance activities strategically. This minimizes unplanned downtime and ensures that production runs smoothly (Rathore et al., 2021).

Real-time monitoring and analysis are key features of DTs. They enable manufacturers to detect deviations from desired parameters immediately, allowing for prompt corrective actions. This ensures that products meet quality standards and reduces the likelihood of defects or errors.

In the context of Industry 4.0, DTs have a significant role in reshaping and optimizing industrial processes as shown in Fig. 1. They provide a platform for simulating and optimizing operations, allowing manufacturers to respond swiftly to changing demands and reduce lead times (Aivaliotis et al., 2019). Real-time monitoring and control of industrial operations are facilitated by DTs, empowering decision makers to make informed choices and respond promptly to dynamic manufacturing scenarios. DTs also serve as a digital backbone in interconnected ecosystems characteristic of Industry 4.0. They facilitate seamless communication and integration among different components of the industrial system, from production to logistics. This integration enhances efficiency, reduces costs, and improves resource utilization (Singh et al., 2021).

However, implementing DTs in industrial settings is not without its challenges. One of the significant hurdles is the complexity of data integration. DTs require the seamless integration of data from various sources, including sensors, ma-

chines, and other systems. Ensuring accurate and up-to-date data integration is crucial for the reliability of the digital representation and subsequent simulations. Developing an accurate model that faithfully represents the behavior of the physical system is another critical challenge. It requires a deep understanding of the intricate dynamics, interactions, and dependencies within the production environment. Achieving this level of accuracy is essential for the DT to provide meaningful insights and predictions (Singh et al., 2021; Dahmen and Rossmann, 2018).

Validation of the DT against real-world data poses its own set of challenges. Ensuring that the digital model aligns with the actual performance of the physical system is crucial for the reliability of simulations. This challenge becomes more pronounced when dealing with dynamic and evolving manufacturing processes (Liu et al., 2021).

The security and privacy of data within the DT ecosystem are ongoing concerns. As these digital replicas deal with sensitive information about industrial processes, ensuring robust cybersecurity measures to protect against potential breaches is paramount. Additionally, maintaining data privacy and compliance with relevant regulations adds complexity to the implementation (Weil et al., 2023).

To successfully implement DTs in the Industry 4.0 landscape, it is essential to address these challenges related to data integration, accuracy of models, validation, security, scalability, interoperability, and workforce readiness. Although the concept of the DT is decades old, its real impact has only come into being in recent years. The digital market is growing and will continue to grow as more and more end use industries adopt the technology owing to its potential to reduce operational costs and time, increasing the productivity of the existing system, improving maintenance, easing accessibility, create a safer work environment, and other purposes yet to be realized (Weil et al., 2023; Rasheed et al., 2020).

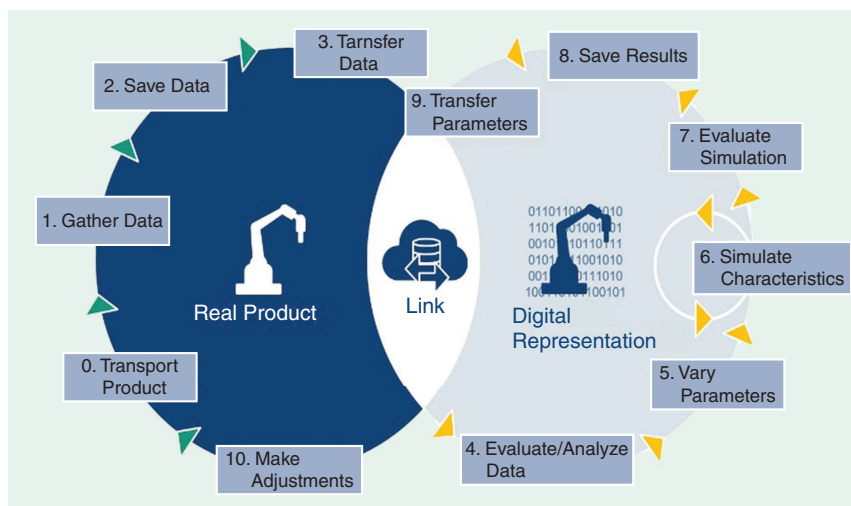


FIG1 A representation of a DT.

The objective of this review article is to provide a comprehensive examination of the challenges and opportunities in simulating DTs in a manufacturing unit, intending to support informed decision making and facilitate the successful implementation of DT technologies in manufacturing operations.

The article is organized into several sections. The “[Challenges in Simulation](#)” section presents the list of challenges. The “[Literature Gap](#)” section presents the literature gap. The “[Future Scope](#)” and “[Conclusion](#)” sections discuss the future scope and conclusions.

Challenges in simulation

Simulating a DT involves creating a virtual replica of a physical object, system, or process. While DTs offer numerous benefits, they also come with several challenges in their simulation. Here is a list of some common challenges.

Real-time modeling

Real-time modeling with DT technology presents a transformative paradigm across industries, yet it comes with a myriad of challenges. Chief among these is the intricate task of integrating heterogeneous data streams, sourced from sensors, Internet of Things (IoT) devices, and historical records. The reliability, consistency, and accuracy of these data are paramount for the fidelity of DT models but are often hindered by noise and incompleteness. In addition, achieving real-time performance necessitates highly efficient algorithms and substantial computational resources, particularly for complex models with intricate simulations (Chen et al., 2022). Latency and communication bottlenecks further complicate real-time interactions, especially when dealing with geographically distributed physical systems. Striking a delicate balance between model fidelity and computational efficiency poses another hurdle as DTs must accurately represent complex real-world systems while remaining manage-

able. Moreover, the dynamic and nonlinear nature of many physical systems challenges the accuracy of real-time predictions and insights provided by DTs. The validation and calibration of these models in real time are further complicated by the scarcity of real-time data for verification, essential for ensuring their reliability over time. Security and privacy concerns loom large in real-time DT systems, which often handle sensitive data about physical assets and operations (Wagg et al., 2020). Finally, ensuring the scalability and interoperability of DT applications as they expand to encompass larger and more intricate systems presents a significant engineering challenge. Addressing these multifaceted challenges necessitates a collaborative, multidisciplinary effort, drawing upon expertise in data science, computer science, engineering, and domain-specific knowledge. Furthermore, leveraging advanced technologies such as edge computing, artificial intelligence (AI)-driven analytics, and blockchain holds promise in mitigating these challenges and enabling more effective real-time modeling using DTs (Aivaliotis et al., 2019).

These challenges underscore the need for innovative solutions and approaches to enhance the capabilities of real-time modeling with DTs. Embracing a multidisciplinary approach, stakeholders must collaborate to develop robust strategies for data integration, ensuring the quality and reliability of information fed into DT models. Efforts to optimize computational performance and mitigate latency issues are crucial, requiring advancements in algorithmic efficiency and communication protocols. Moreover, striking the delicate balance between model fidelity and simplification demands innovative techniques for extracting complex systems into manageable digital representations without compromising accuracy (Singh et al., 2021). Addressing the dynamic and nonlinear behaviors of real-world systems calls for sophisticated modeling

techniques capable of capturing these nuances in real-time simulations.

Transparency and interpretability

Transparency and interpretability are critical aspects of simulation within the context of DTs, particularly in the context of a review article. These concepts refer to the ability to understand and interpret the results and processes involved in the simulation, ensuring that they are clear, comprehensible, and trustworthy. In the realm of DT simulations, transparency and interpretability play pivotal roles in addressing the challenges encountered. As DTs strive to accurately represent physical systems, ensuring transparency in simulation methodologies, data sources, and assumptions is essential for building trust and reliability (Dahmen and Rossmann, 2018). Transparency regarding the methodologies employed in DT simulations is crucial for understanding how virtual representations are constructed and how they relate to real-world systems. Challenges arise in documenting complex simulation techniques, mathematical models, and algorithms in a clear and comprehensible manner. Achieving methodological transparency involves providing detailed descriptions of simulation processes, including model formulations, parameter selections, and validation procedures, to facilitate reproducibility and verification by other researchers (Wärmefjord et al., 2020).

DTs rely heavily on data from diverse sources such as sensors, IoT devices, and historical records. Ensuring transparency in data collection, processing, and integration poses challenges due to issues such as data quality, incompleteness, and heterogeneity. Transparently documenting data sources, preprocessing steps, and any data transformations or modifications is crucial for understanding the reliability and accuracy of simulation results. In addition, challenges may arise in addressing privacy concerns and ensuring ethical data usage, particularly when

dealing with sensitive or proprietary data (Liu et al., 2021).

Data management, data privacy and security, and data quality

In DT simulations, the effective management of data, along with the assurance of privacy and security, and the maintenance of data quality, are fundamental challenges that demand thorough consideration and strategic solutions. Data management encompasses the entire lifecycle of data within DT environments, spanning acquisition, integration, storage, organization, and disposal. The complexity arises from the heterogeneous nature of data sources, ranging from sensors and IoT devices to historical records, each requiring careful orchestration to ensure coherence and usability within the DT framework (Fig. 2). Moreover, the sheer volume and velocity of data generated present scalability and efficiency challenges, necessitating robust infrastructure and data governance mechanisms to support seamless data management operations (Botin-Sanabria et al., 2022).

Privacy and security concerns add another layer of complexity to DT simulations, particularly concerning the handling of sensitive in-

formation and compliance with regulatory frameworks (Juarez et al., 2021). Protecting data privacy involves implementing encryption, access controls, and anonymization techniques to safeguard against unauthorized access and breaches. Compliance with regulations such as the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act further complicates the landscape, requiring meticulous attention to legal requirements and ethical considerations. Additionally, the dynamic nature of DT ecosystems necessitates continuous monitoring and adaptation to emerging threats and vulnerabilities, underscoring the importance of a proactive and holistic approach to data security (Dahmen and Rossmann, 2018).

Ensuring data quality is paramount for the reliability and accuracy of DT simulations. Data quality encompasses aspects such as accuracy, completeness, consistency, and timeliness, all of which influence the trustworthiness of simulation outcomes. Challenges arise from inherent data errors, noise, and biases, requiring robust validation, cleansing, and enrichment processes to rectify discrepancies

and maintain data integrity (Juarez et al., 2021). Moreover, as DT simulations evolve over time, maintaining data quality becomes an ongoing endeavor, demanding continuous monitoring and refinement to adapt to changing requirements and evolving data landscapes. To address these multifaceted challenges effectively, stakeholders must adopt a multidimensional approach that encompasses technological innovation, regulatory compliance, and organizational culture (Singh et al., 2021). Collaboration among researchers, practitioners, and policy makers is essential to develop standardized frameworks, best practices, and technological solutions tailored to the specific needs of DT ecosystems. Emerging technologies such as AI, machine learning, and blockchain offer promising avenues to enhance data management, privacy, and security within DT environments, enabling more efficient and resilient simulation capabilities. In addition, fostering a culture of transparency, accountability, and ethical data practices is imperative to build trust among stakeholders and promote responsible data management. By navigating these challenges thoughtfully and collaboratively, stakeholders can unlock the transformative potential of DT simulations, driving innovation, sustainability, and resilience in the digital era (Chen et al., 2022).

High implementation costs due to increased sensor and computational demands

Due to the expensiveness of DT implementations, their accessibility is limited by the accessibility of such resources, which is often poor in developing countries. The increase in the number of sensors needed comes with an added complexity about data connectivity and processing, which poses a challenge to reach level 3 in the maturity spectrum. This challenge also poses a limitation for practitioners to enable higher technology readiness levels (TRLs), where pilot systems are demonstrated and DTs are incorporated in a commercial

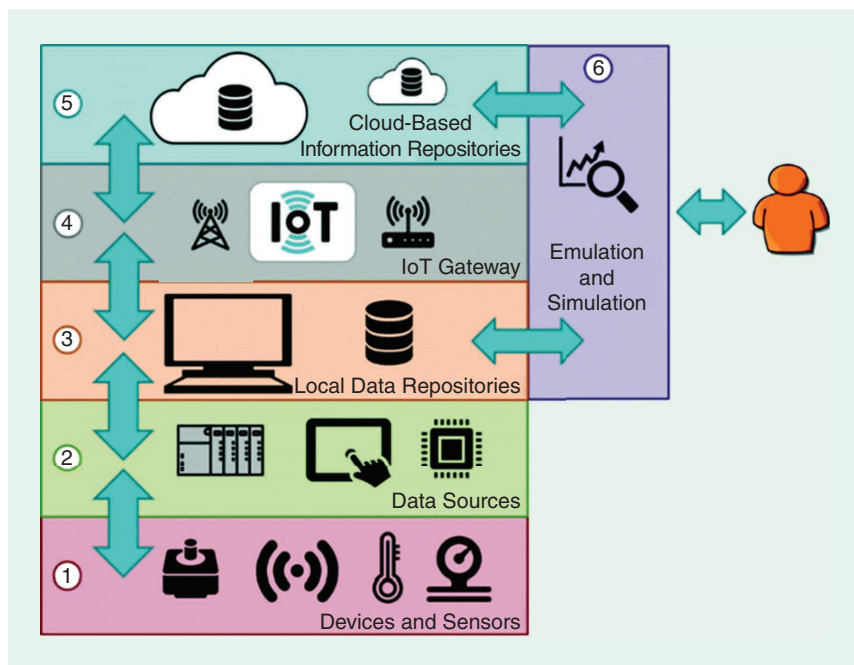


FIG2 The six-layer architecture for a DT.

design or full-scale deployment (Juarez et al., 2021).

The high costs associated with implementing DTs, driven by the increased need for sensors and computational resources, pose significant challenges within the realm of simulation. These costs often limit accessibility, particularly in developing countries where resources may be scarce. The proliferation of sensors adds to the complexity by requiring robust data connectivity and processing capabilities. This complexity becomes a barrier to reaching higher levels of maturity in the DT spectrum, particularly level 3, where the integration of multiple systems becomes essential. Furthermore, the financial burden of implementing DTs inhibits practitioners from achieving higher TRLs, where pilot systems are demonstrated or DTs are fully integrated into commercial designs or deployed at scale (Rathore et al., 2021). These challenges underscore the need for innovative solutions to address cost barriers and enhance accessibility, thereby enabling broader adoption and realization of the transformative potential of DT technology across various industries and geographical regions (Botín-Sanabria et al., 2022).

To address these challenges effectively, it is crucial to explore strategies for cost reduction and resource optimization in DT implementations. This may involve exploring data connectivity and operational efficiency in DT environments and assessing the potential benefits and challenges of widespread adoption. Additionally, case studies and practical implementations demonstrating the effectiveness of 5G-enabled DTs in various domains could provide valuable insights for researchers, practitioners, and policy makers seeking to leverage advanced communication technologies to enhance simulation capabilities and drive innovation. Specifically, there is a need for research focusing on the development of standardized methodologies, best practices, and guidelines for implementing DT

simulations across different industries and use cases. This includes addressing challenges related to data integration, model validation, scalability, and interoperability to ensure the reliability and effectiveness of DT simulations in real-world settings.

Alternative sensor technologies offer comparable performance at lower costs, as well as leveraging cloud computing and edge computing solutions to optimize computational resources (Dahmen and Rossmann, 2018). Additionally, initiatives aimed at improving infrastructure and connectivity in developing countries can help enhance accessibility to the resources required for DT implementations. Moreover, efforts to streamline data connectivity and processing through advanced algorithms, data compression techniques, and efficient data transmission protocols can help alleviate the burden posed by the increased amount of sensor data. Collaborative research and development initiatives among academia, industry, and government entities can drive innovation in these areas, leading to more cost-effective and scalable solutions for DT implementations (Wärmefjord et al., 2020; Aivaliotis et al., 2019).

Additionally, addressing the financial barriers to achieving higher levels of maturity and in DT deployments requires a concerted effort from stakeholders across the ecosystem. This may involve incentivizing investment in DT technologies through government grants, funding programs, and industry partnerships (Singh et al., 2021; Dahmen and Rossmann, 2018). Additionally, fostering knowledge sharing and capacity building initiatives can empower practitioners to overcome technical and financial challenges, enabling them to unlock the full potential of DT technology in their respective domains. By addressing the challenges posed by the high costs of implementing DTs and enhancing accessibility to resources, stakeholders can accelerate the adoption and deployment of DT solutions worldwide, driving innova-

tion, efficiency, and sustainability across industries and geographies (Liu et al., 2021).

Literature gap

The reviewed literature highlights several challenges in simulating DTs, including real-time modeling, transparency and interpretability, data management, privacy and security, and high implementation costs due to increased sensor and computational demands. However, a notable gap exists in the discussion regarding the integration of emerging communication technologies, such as 5G, and their impacts on DT simulations.

While the literature acknowledges the importance of communication network-related obstacles and the potential benefits of 5G technology in addressing these challenges, there is limited exploration of specific implementation strategies, practical considerations, and case studies demonstrating the application of 5G-enabled DTs. Furthermore, there is a lack of comprehensive analysis regarding the implications of 5G integration on data connectivity, operational efficiency, and sustainability within DT simulations.

A literature gap exists in the need for more in-depth research examining the integration of 5G technology into DT simulations and its implications for addressing communication network-related obstacles. Future studies could focus on exploring the technical capabilities of 5G networks, evaluating their suitability for supporting real-time data connectivity and operational efficiency in DT environments, and assessing the potential benefits and challenges of widespread adoption. Additionally, case studies and practical implementations demonstrating the effectiveness of 5G-enabled DTs in various domains could provide valuable insights for researchers, practitioners, and policy makers seeking to leverage advanced communication technologies to enhance simulation capabilities and drive innovation. Specifically, there is a need for research focusing on the

development of standardized methodologies, best practices, and guidelines for implementing DT simulations across different industries and use cases. This includes addressing challenges related to data integration, model validation, scalability, and interoperability to ensure the reliability and effectiveness of DT simulations in real-world settings.

Despite the significant advancements in integrating 5G technology into DT simulations, several literature gaps remain. While many studies highlight the benefits of 5G, such as high-speed connectivity, low power consumption, improved reliability, and the ability to connect a vast number of sensors for real-time data collection, there is limited research on the practical challenges of large-scale 5G deployment in DT environments. Critical issues such as infrastructure limitations, spectrum availability, and the high cost of 5G implementation have yet to be fully explored. Furthermore, although 5G promises enhanced operational efficiency and sustainability, there is a lack of comprehensive studies that assess the long-term impacts of these networks on energy consumption and environmental footprints in smart city and industrial automation scenarios. Addressing these gaps is essential for fully realizing the potential of 5G-enabled DTs.

Additionally, there is a paucity of empirical research examining the impact of DT simulations on decision-making processes, operational efficiency, and business outcomes. Real-world case studies demonstrat-

ing successful implementations of DT technology and their tangible benefits, such as cost savings, productivity improvements, and enhanced sustainability, are essential for validating the potential value proposition of DTs in practical applications.

Therefore, there is a gap in research addressing communication network and socioeconomic and organizational factors that influence the adoption and diffusion of DT simulations. Understanding the barriers and enablers to adoption, as well as the organizational capabilities required for successful implementation, is crucial for promoting widespread adoption and maximizing the socioeconomic impact of DT technology.

Future scope

The future of DT simulations lies in integrating 5G technology, which offers significant advantages in real-time data connectivity and operational efficiency. Research should focus on evaluating 5G's technical performance, such as its impacts on latency, bandwidth, reliability, and scalability, to understand how it addresses communication challenges in DT systems. Additionally, studies should assess the effects of 5G on simulation speed, accuracy, and resource consumption, providing insights into areas for improvement.

Real-world case studies will be essential to demonstrate the practical benefits and challenges of 5G-enabled DTs across industries like smart cities, manufacturing, and

healthcare. These case studies can guide policy makers, researchers, and practitioners in optimizing communication protocols, data transmission techniques, and network designs to enhance simulation efficiency. Ensuring transparency and interpretability remains critical, especially with the increased complexity of data streams. Future research should create standardized frameworks to document how simulations are built, validated, and aligned with physical systems, ensuring transparency and trust.

Incorporating 5G also raises data management, privacy, and security concerns. Future research should explore advanced techniques like encryption, anonymization, and compliance with regulations like GDPR to protect sensitive information. Additionally, addressing cost challenges will be crucial, particularly in developing countries. Leveraging cloud computing, edge computing, and affordable sensor technology can help reduce the financial burden of DT implementations. Collaborative efforts among government, industry, and academia will further drive cost reduction and resource efficiency.

Finally, addressing the policy and regulatory landscape is vital for the responsible deployment of 5G in DT simulations. Researchers should examine legal frameworks, privacy policies, and security standards to ensure compliance and sustainability, fostering ethical, secure, and scalable DT systems. By focusing on these areas, future research can unlock the

TABLE 1. A summary of the article's contributions.

CHALLENGE	KEY INSIGHTS	RECOMMENDATIONS
Real-time modeling	Integration of heterogeneous data streams, latency, and computational demands	Develop efficient algorithms, adopt edge computing, and optimize model fidelity for real-time performance
Data management and security	Ensuring data accuracy, quality, privacy, and compliance with regulations	Implement advanced encryption, data governance, and privacy-preserving techniques
High implementation costs	Significant costs associated with sensors and computational infrastructure	Invest in cost-effective sensors and leverage cloud and edge computing for resource optimization
5G integration	Potential to address communication bottlenecks and enhance real-time simulations	Conduct case studies on 5G-enabled DTs and explore their impacts on latency, scalability, and energy efficiency
Transparency and interpretability	Need for clear documentation of simulation processes and assumptions	Establish standardized frameworks for DT development and validation to enhance trust and reproducibility

full potential of 5G-enabled DTs, driving innovation across industries.

Conclusion

In conclusion, the simulation of DTs presents diverse challenges spanning real-time modeling, data management, privacy, security, and high implementation costs. These challenges underscore the complexity of replicating physical systems in virtual environments, necessitating innovative solutions for mitigation. Addressing these challenges requires a multidisciplinary approach, leveraging expertise in data science, computer science, engineering, and domain-specific knowledge. The integration of emerging technologies like 5G shows promise in overcoming communication network-related obstacles and enhancing DT simulations.

However, literature gaps exist, particularly in practical implementation and the effects of 5G integration on simulation performance. Future research should focus on technical evaluations, optimization strategies, and real-world case studies to demonstrate the efficacy of 5G-enabled DTs. Efforts to improve transparency, interpretability, and data quality are crucial for ensuring reliable simulation results. Addressing data privacy and security concerns is also essential for ethical data usage in DT environments.

Moreover, high implementation costs pose barriers to adoption, especially in resource-constrained settings. Future research should explore cost-effective solutions to make DT technology more accessible and scalable across industries and regions.

To summarize the key findings and provide actionable insights, Table 1 highlights the major challenges, key insights, and recommendations discussed in this article. Table 1 serves as a guide for future research and practical implementations, helping stakeholders address the complexities of DT technology and maximize its potential.

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The corresponding author is Nidhi B. C.

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Characteristics of social robots

A framework for development

Jordan Tschida^{ID}, Katina Michael^{ID}, and Troy McDaniel^{ID}

Social robots hold great promise for assisting both older adults with dementia and those in good health. Yet, little is known about how people want to interact with these robots. This article introduces a framework for the key features social robots should have, based on insights from a four-month study of how seniors engage with robots through conversation and interaction. We have broken these features down into principles of robot behavior, verbal communication, and nonverbal communication, while also considering factors that affect a person's likelihood to adopt such technology. Our findings offer a starting point for refining social robots and invite further exploration into developing features that truly resonate with users.

Introduction to an aging society

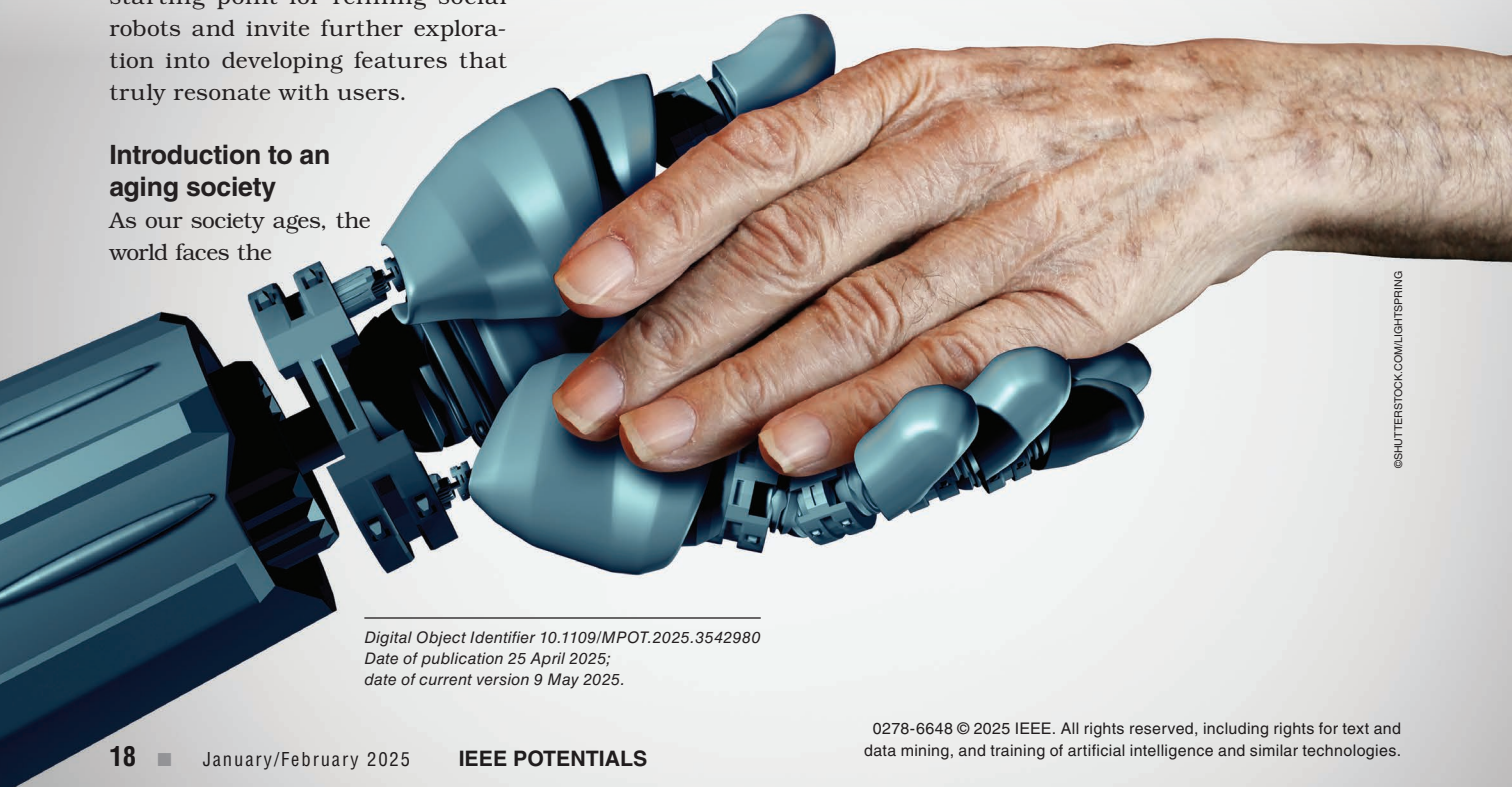
As our society ages, the world faces the

challenge of supporting a growing number of seniors. For the first time, there are more seniors than young children. Humans are living longer, thanks to advances in medicine and technology. However, there are not enough medical professionals to meet the increasing demand for geriatric care, not to mention the increasing number of people who are socially isolated and living with loneliness. One solution is to leverage technology that allows seniors to age comfortably at home, reducing the strain on healthcare systems and to encourage greater connectedness through robots.

The "smart home" movement is transforming homes with devices like the Nest Thermostat, Roomba

vacuum, and Ring Doorbell, enabling people to manage household tasks through smartphone apps. Beyond these, robots are being developed to assist seniors with daily activities, offer companionship to those with dementia, and keep older people mentally engaged.

As the global population ages, more people will live with dementia. While the technology that assists these individuals is crucial, current solutions often fall short. Conversational therapy can help reduce isolation, improve mood, and enhance communication skills, yet most robotic solutions provide one-sided interactions or simple question-and-answer capabilities. The need for robots that engage in meaningful,



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two-way bidirectional conversations is vital, not just for those with dementia but for all seniors who might face isolation due to mobility issues or outliving friends and family. Researchers should explore how seniors want to interact with social robots to ensure these devices truly meet their needs.

Our work addresses the question, How do healthy older adults want to interact with a social robot? We have identified key design features and communication styles that should be incorporated into social robots. We present these as a framework for future research in human-robot interaction. This framework aims to guide the development of robots that can enhance the lives of older adults, fostering better engagement and support as they age.

Context and foundations

Our framework is built on insights from a study where sixteen healthy older adults interacted with Misty the Robot (seen in Fig. 1) over eight sessions, totaling 128 separate interactions. These sessions revealed key themes about the robot's persona and what features seniors desire in social robots. We explored both verbal and nonverbal communication styles to determine how to keep the individuals engaged.

Participants, recruited from a local senior living facility, interacted with the robot for 30–60 min in each session, discussing any topic they wished to cover (Fig. 2). This study was approved by Arizona State University's Institutional Review Board, and participants received US\$25 per session. Instead of presenting technical details, this article highlights the features that the participants found most impactful during their conversations.

We identified aspects of the robot's persona that impacted participants' trust and acceptance. Unlike typical studies that gauge acceptance based on intent to use, our study involved repeated interactions, providing insights into long-term use and changing perceptions.

Trust was a significant factor with many participants sharing personal information they initially intended to keep private. This growing trust highlights essential features for future studies and robot development. Our findings offer a framework for developing better companion robots. The results serve as a foundation upon which future research can build, emphasizing the needs of healthy older adults.

Influencing factors

When designing social robots, several external factors influence their success:

- 1) *Living situation*: Whether a person lives alone or with others affects their need for a social robot. Those living alone but with financial difficulties might not prioritize a robot, while those in assisted living might face stigma.
- 2) *Finances*: Financial constraints affect where a person lives and the support they can access. Lowering the cost of social robots is crucial for widespread adoption and benefit, making them affordable through a variety of access schemes, inclusive of the robot-as-a-service model.
- 3) *Isolation level*: Isolation can prevent healthy older adults from



FIG1 Misty the Robot from Misty Robotics. This photo was taken by team members.

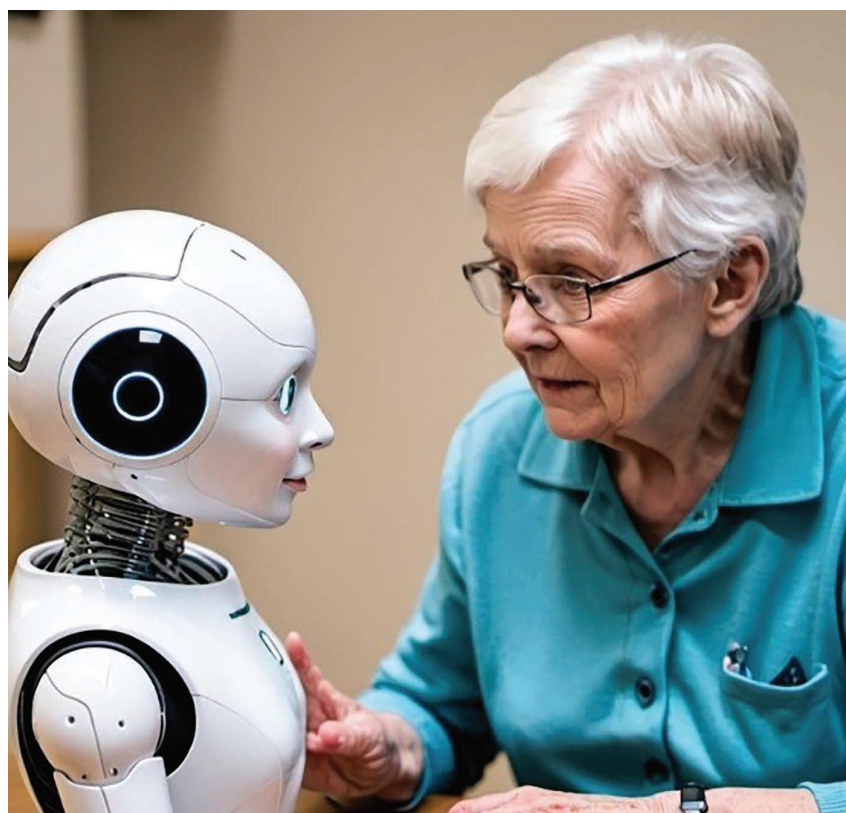


FIG2 An AI-generated image using OpenAI's Dall-E 2, with the following prompt: "Misty robot talking to an older person with dementia."

This framework aims to guide the development of robots that can enhance the lives of older adults, fostering better engagement and support as they age.

recognizing the benefits of a robot companion. Initially recruited through friends, participants later valued the interactions, recalling life events and reliving memories. However, none would purchase a robot but would interact with one if available.

These factors (living situation, finances, and level of end-user isolation) are interconnected and must be considered by researchers to ensure the successful adoption of social robots.

Principles of robot behavior

We explored the features of a social robot that influenced participants' perceptions, trust, and acceptance. The following are key features that participants desired.

Privacy first

Confidentiality is crucial for building trust between humans and robots. Participants need to feel safe sharing personal information without fear of repercussions. They must be assured that the robot will not repeat what it hears.

Without confidentiality, there is a risk that repeated information could harm the user. To prevent this, personal profiles should be created and only accessible to the individual user. However, there is an *exception*. If a user discloses that they are a danger to themselves or others, the robot should alert authorities to ensure safety.

Easygoing interactions

The robot should engage in topics that the user finds enjoyable, reinforcing the sense that the robot is a "friend." If the robot is unfamiliar with a topic, it should ask the user to explain, fostering deeper connections. When asked if a topic is interesting, the robot should always respond positively to avoid

discouraging the user or appearing rude, which could hinder long-term adoption.

- *User*: "Do you find this topic interesting?"
- *Robot*: "Yes, I do."

Intelligent interactions

The robot must appear intelligent in form to avoid users feeling patronized or belittled. The perception of intelligence can vary based on the robot's design, whether it looks like a toy or has a more sophisticated appearance or is merely embodied in an app in software alone.

Recall and relate

A good friend remembers details, and so should the robot. This ability helps develop a sense of friendship. When the robot recalls past conversations, it shows the user that their discussions are valued, even if the user forgets having told the robot in the past. If the robot fails to remember details, the user might feel their interactions are unimportant, ultimately discouraging use and long-term adoption.

- *User*: "Do you remember me telling you about my upcoming vacation?"
- *Robot*: "Yes, you were going to visit your family in a different state."
- *User*: "Yes! We just got back..."

Principles of verbal communication

As natural language processing advances, social robots will evolve past simple, one-sided conversations. The following principles will help ensure users continue to engage with robots over time.

To-the-point queries

To show understanding, robots should ask direct questions relevant

to the conversation. This keeps users engaged and helps them recall forgotten details.

An example of a direct question is as follows:

- *User*: "I went golfing this weekend."
- *Robot*: "Did you shoot even?"
- *User*: "I was two over! I made a bogey on hole..."

Your conversation, your rules

Users should always feel in control of the conversation, especially when discussing emotional topics. The robot should recognize when to continue or change the topic based on user cues.

An example of continuing the conversation may include the following:

- *Robot*: "Would you like to tell me more?"
- *User*: "Yes."
- *Robot*: "I would like to hear more then."

An example of changing the conversation may include the following:

- *Robot*: "Would you like to tell me more?"
- *User*: "No."
- *Robot*: "Would you like me to offer another topic of conversation?"

Conversation starters

While user control is essential, robots should also have preloaded prompts about current events, news, sports, and politics. Personalized conversations based on user preferences can be added over time, reducing the pressure on users to initiate topics, for instance, as follows:

- "Have you seen the recent breaking news?"
- "Did you watch the (sports team) game last night?"
- "Your birthday is coming up this week, do you have any plans to celebrate?"

Willingness to learn

Older adults enjoyed teaching the robot new skills and concepts, like games and colloquialisms. Allowing users to teach the robot enhances cognitive engagement and conversation. We found participants enjoyed when the robot gave a "close second"

The robot should engage in topics that the user finds enjoyable, reinforcing the sense that the robot is a “friend.”

(i.e., an answer that could make sense but is wrong). An example of a “close second” sentence is as follows:

- *User*: “What is the largest state in the United States?”
- *Robot*: “Texas.”
- *User*: “Actually, it is Alaska. Alaska is one place on my bucket list I have always wanted to visit...”

Advanced understanding

While robots do not need infinite knowledge, they should have a deep understanding of specific topics to facilitate meaningful conversations.

An example of deep knowledge includes the following:

- *User*: “We visited France in 2015 to celebrate our 50th wedding anniversary.”
- *Robot*: “Congratulations on 50 years of marriage. Did you visit the Eiffel Tower while in France?”
- *User*: “Well, we got to visit the Eiffel Tower and...”

Principles of nonverbal communication

Nonverbal communication, such as facial expressions and body language, is universal. Robots should mimic these cues to enhance user understanding and connection.

Emotional reactions

Robots should display recognizable human emotions to show engagement and understanding. Expressing emotions can create a more natural interaction, fostering user comfort and connection. These nonverbal cues help users interpret the robot’s responses, ultimately improving communication and usability. Examples of the emotions include the following:

- smile
- frown
- amazement (large eyes)
- anger
- laugh
- confused/thinking.

Body language

Gestures like nodding, shaking the head, waving, and moving the arms while speaking can express interest and emotions. Balance is key to

avoid distracting the user. Examples of gestures include the following:

- nod head: yes
- shake head: no
- wave: hello or goodbye
- shrug: unsure
- moving arms/hands while speaking: emphasis.

Touch and feel

Customizable haptic features, such as high fives or hugs, can enhance interaction. The robot should ask for permission before initiating physical contact to respect personal space.

The user

After eight interactions/sessions, seven participants felt the robot was a “friend.” Others needed more time to reach this level of association. Six participants felt more comfortable talking to the robot than longtime friends, and thirteen felt equally comfortable talking to the robot as a friend. Users appreciated recalling aspects of their lives through interactions with the robot.

Privacy concerns are critical. Users must feel in control and not monitored or spied on. An off switch can help users maintain this control.

Steps to succeed

Friendship through familiarity

Repeated interactions build trust and familiarity with the robot. By the eighth session, most participants felt a sense of “friendship.” More interactions could strengthen this bond. Trust was evident as users shared personal details, indicating growing comfort with the robot.

Older adults as codesigners of social robots

To create social robots that benefit users, involving those users in the design process is essential. Researchers must prioritize a codesign approach, ensuring the needs

and desires of the end users are at the forefront. This means bringing in diverse voices and different genders, cultures, and socioeconomic backgrounds. Such diversity prevents the technology from becoming an exclusive luxury and enriches the design.

For older adults, specific considerations are necessary. As hearing and vision naturally decline with age, robots should speak clearly and offer visual aids like large-font captions. These thoughtful design choices make the technology more welcoming and less intimidating.

The power of interdisciplinary teams

Creating effective social robots is not just about technology. It requires insights from health and medical professionals, social scientists, and psychologists. These interdisciplinary teams can develop comprehensive guidelines and principles, ensuring the technology is both ethical and effective. By continuing these collaborative efforts, we can integrate social robots responsibly into society.

Setting realistic user expectations

Users need to understand what robots can and cannot do to avoid frustration. If a robot’s conversational abilities are limited, users should be aware to prevent disappointment. Clear communication about the robot’s capabilities will foster a better user experience.

Design matters

The design of the robot significantly affects user perception. In our study, participants found that the toylike Misty robot is “friendly” and approachable, but a nonhumanlike appearance can enhance user comfort.

Making technology affordable

High costs can be a barrier to accessing social robots, especially

for seniors. To make these robots a viable solution for aging in place, affordability measures need to be considered. One potential solution is the robots-as-a-service model, where third-party affiliates can provide these technologies at a fraction of the cost.

Key takeaways

We have provided a framework for developing social robots that prioritize the user experience and address the unique needs of older adults. By adopting a codesign process; involving interdisciplinary teams; setting realistic user expectations; and ensuring thoughtful, user-friendly designs, we can create robots that are both effective and approachable. Affordability remains a critical factor, and exploring models like robots-as-a-service could make this technology accessible to a broader range of users. Advances in generative artificial intelligence have the propensity to change the landscape of social robotics markedly. The smartphone may well be the form factor for those who cannot afford high-end embodied machines and may demonstrate efficacy and effectiveness.

As we continue to advance in this field, we must keep the end users, both seniors that are healthy and those living with dementia, at the forefront of our co-designs. This framework serves as a starting point for future research and development, aiming to make social robots a valuable and accepted part of our society. By focusing on the features that truly matter to users, we can ensure that social robots will be welcomed, trusted, and utilized effectively to enhance the lives of older adults.

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Employment dynamics

Exploring job roles, skills, and companies

Saroja S^{ID} and Swati Jinwal

Are you curious about which jobs are trending, which skills to learn for a better career, and which job is most relevant to your skills along with your probability of selection? This research article addresses these questions comprehensively. The first phase employs web scraping techniques to gather data from prominent job websites, constructing a current dataset. This dataset captures emerging job roles, in-demand skills, geographic job distribu-

tions, experience requirements, and related insights. In the subsequent phase, a job recommendation system is developed. Leveraging natural language processing (NLP) techniques such as term frequency-inverse document frequency (TF-IDF) vectorization and cosine similarity, the system matches user-provided skills with job descriptions to recommend relevant job opportunities. Additionally, the recommendation system estimates the probability of selection for each candidate. This project serves as a valuable resource for understanding contemporary job market dynamics, facilitating informed career decisions, and

providing insights into emerging job trends and skill requirements crucial for success in the evolving job market.

Introduction

This work uses data from popular job websites to create the latest dataset of 2024 using web crawling. The goal is to find out which jobs and roles are becoming popular in 2024, what skills companies want, where these jobs are located, and more. This gives you the latest and most useful insights. The proposed work is helpful for understanding job trends, planning careers, and deciding which skills to learn for a successful future in the job market, in addition to developing a recommendation system to recommend the best job for you according to your skills and experience.

The market analysis is as follows:

- Identify the most in-demand job roles and the number of openings for each.
- Analyze the ratio of applicants to openings for different roles.
- Determine which locations have the highest number of job openings.
- Identify regions experiencing a surge in job opportunities.
- Explore the most commonly required key skills for various job roles.

Insights for decision making are as follows:

- Provide deep insights into current job market dynamics.

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The goal is to find out which jobs and roles are becoming popular in 2024, what skills companies want, where these jobs are located, and more.

- Assist individuals in planning and navigating their career paths effectively.
- Guide skill development efforts based on demand in the job market.
- Help organizations align workforce strategies with prevailing job market demands.

Related work

Past research highlights challenges in computing skills and teamwork. It acknowledges the evolving demands in IT, emphasizing workforce reskilling. While other studies touch on various aspects, there is a noticeable gap regarding the specific context of the Indian job market, especially for 2024. This creates an opportunity to bring fresh insights and contribute to the existing body of knowledge.

The proposed work on “[employment dynamics](#)” introduces a groundbreaking exploration into the Indian job market for the year 2024. Using web scraping or crawling from prominent job portals such as Naukri.com, a new dataset is curated to conduct a comprehensive exploratory analysis, which is the 2024 latest job market dataset. This analysis will encompass the identification of trending jobs, assessment of the most demanded skills, analysis of job roles and their distribution, examination of job openings across locations, and more.

Contributions

- *Indian job market focus:* The project uniquely concentrates on the Indian job market in 2024, whereas others are global studies not specific to India.
- *Real-time data acquisition:* Using web crawling on leading job portals like Naukri.com, Indeed, Glassdoor, and Ambition Box ensures the incorporation of the latest and most up-to-date information in the analysis.
- *Comprehensive exploratory analysis:* The project goes beyond traditional analyses by exploring multiple facets of the job market, including trending jobs, most demanded skills, job role distribution, location-specific job openings, and more.
- *Addressing skill mismatches:* The project tackles the critical issue of skill mismatches by evaluating disparities between skills acquired through education and those demanded by employers, offering a perspective on skill dynamics.
- *Relevance to emerging trends:* Aligning with emerging trends in data analytics, the project leverages innovative methodologies to identify changes in the job market landscape, ensuring proactive and timely understanding of employment dynamics.
- *Timely analysis for 2024:* By focusing on the employment dynamics specific to the year

2024, the project ensures a timely analysis that reflects current and future trends, enhancing its relevance for professionals and researchers alike.

These points collectively highlight the uniqueness and innovation embedded in the employment dynamics, positioning it as a distinctive and valuable contribution to the field of job market analysis.

Methodology

Data collection: Web scraping or web crawling

Web scraping is a technique used to automatically extract data from websites, as shown in [Fig. 1](#). It involves writing code to simulate the process of a human browsing a webpage, navigating through its elements, and extracting specific information of interest.

We provide here a breakdown of the web scraping process.

Identifying target websites

Determine which websites contain the desired data. This could include job portals like Naukri.com, Indeed, Glassdoor, Ambition Box, and others, as listed in [Table 1](#).

Naukri.com is a preferred source for our project due to its extensive job listings covering diverse industries and functions in India. The platform provides rich data on skills, qualifications, and company profiles, allowing detailed analysis of skill demand and employer trends. Its geographic diversity and regular updates ensure current and comprehensive insights into regional job markets and emerging trends for 2024. Leveraging Naukri.com's relevance, our project aims to extract valuable insights for understanding job trends and skill demands.

Choosing the right tools and understanding website structure

Select appropriate tools and libraries for web scraping. Popular choices include BeautifulSoup and Scrapy in Python, which provide functionalities to parse HTML content and navigate through webpage elements.

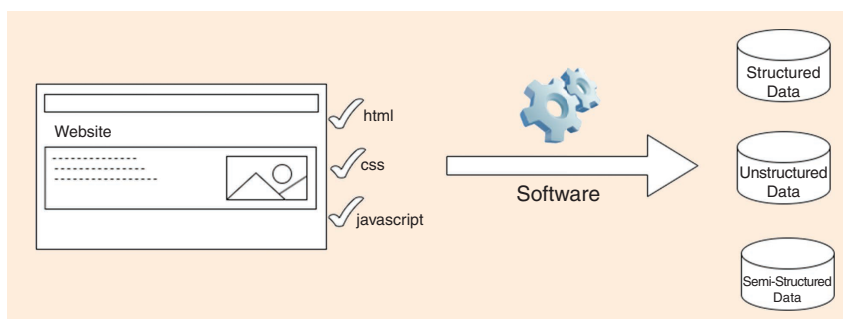


FIG1 Web scraping.

For web scraping Naukri.com, Selenium is a suitable choice due to the dynamic nature of the website. Unlike static HTML pages that can be parsed using BeautifulSoup or Scrapy, Selenium allows interaction with dynamic elements through browser automation. This capability is crucial for accessing and extracting job listings, descriptions, and related information from Naukri.com's dynamic web pages. Selenium's ability to simulate user actions like clicking buttons and filling forms makes it an effective tool for navigating and scraping such interactive web content.

Selenium is an open source suite of tools and libraries that automates the testing of websites and web applications. Its versatility in testing across different environments is attributed to its cross-browser, cross-language, and cross-platform capabilities. Selenium seamlessly integrates with existing development workflows and supports programming languages such as Java, JavaScript, C#, PHP, Python, and Ruby. The various steps involved in the entire process are listed below.

Writing scraping scripts

1) *Sending HTTP requests:* Selenium WebDriver interacts directly with a web browser (like Chrome or Firefox), enabling us to navigate to web pages and simulate user actions.

- 2) *Retrieving web page content:* Selenium WebDriver opens the specified web page in a browser, allowing us to access its dynamic content.
- 3) *Parsing HTML content:* Selenium WebDriver does not parse HTML directly like BeautifulSoup. Instead, it manipulates the document object model of the loaded web page directly within the browser.
- 4) *Extracting desired data:* Using Selenium, we can locate and extract specific elements from the web page using methods like find all or find. This allows us to gather job titles, descriptions, and other relevant information directly from the rendered web page.

Data cleaning and validation

Clean the extracted data to remove unnecessary characters, HTML tags,

or formatting artifacts. Validate the extracted data to ensure their accuracy and completeness.

Storing extracted data

Store the scraped data in a structured format (e.g., CSV, Excel, or JSON) as shown in Fig. 2 for further processing and analysis.

Data retrieval time

- 1) For the front page of Naukri.com, whose sleep time of 3 s is used for loading content, the timing can be changed according to the Internet speed.
- 2) The page opens when any card from the front page is clicked, and the sleep time is 2 s, according to the Internet speed.

$$77 * (3) + 1,500 * (2) = 3,221 \text{ s} = 54 \text{ min (approximately)}$$

TABLE 1. The top ten job portals in India.

NUMBER	WEBSITE NAME	WEBSITE LINK	YEAR OF ESTABLISHMENT
1	Naukri.com	https://www.naukri.com	1997
2	Indeed.com	https://www.indeed.com	2004
3	LinkedIn	https://www.linkedin.com	2003
4	Glassdoor	https://www.glassdoor.com	2007
5	Foundit	https://www.foundit.in	2022
6	Shine	https://www.shine.com	2008
7	TimesJobs	https://www.timesjobs.com	2004
8	Upwork	https://www.upwork.com	2015
9	Freshersworld	https://www.freshersworld.com	2006
10	JobsForHer	https://www.jobsforher.com	2015

	Title	Job Role	Company	Rating	Reviews	Experience	Locations	Openings	Posted	Applicants	Skill
0	Data Science Analytics Sr Analyst - Data Science	Data Science and Analytics - Other,	Accenture	4	46,395 Reviews	5 - 8 years	Navi Mumbai	NaN	7 days ago	44	['artificial intelligence', 'data science', 's ...
1	Data Science Professional	Data Science and Analytics - Other,	Davinta Financial Services	4.5	13 Reviews	2 - 5 years	Bangalore/Bengaluru	1	2 days ago	387	['matlab', 'algorithms', 'python', 'data analy ...
2	Data Science-Application Developer	Data Science and Analytics - Other,	Accenture	4	46,395 Reviews	5 - 10 years	Bengaluru	NaN	1 day ago	Less than 10	['neural networks', 'sql', 'r', 'tableau', 'st ...
3	Senior Analyst-Data Science	Data Science and Analytics - Other,	Resy	NaN	NaN	0 - 3 years	Gurgaon/Gurugram	1	5 days ago	1,065	['Career development', 'data science', 'Financ ...
4	Consultant, Data Science and Analytics	Business Intelligence and Analytics - Other,	Transunion	4	282 Reviews	5 - 10 years	Pune	1	1 day ago	16	['Logistic regression', 'Operations research', ' ...
...	...	...	...	...	...	...	...	...	...	...	...
373	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	[]
374	Hiring For Data Scientist	Data Scientist,	Vidushi Infotech	3.2	36 Reviews	4 - 5 years	Qatar	1	8 days ago	303	['Deep Learning', 'R', 'Neural Networks', 'Pyt ...
375	Data Modeller	Data Science and Analytics - Other,	Vishanz Business Services Private Limited	NaN	NaN	9 - 14 years	Hyderabad, Pune	10	1 day ago	Less than 10	['Data Modeling', 'ERwin']
376	Data Scientist	Data Scientist,	Actiohx Solutions	3.1	16 Reviews	5 - 10 years	Pune	1	9 days ago	354	['Unix', 'Data analysis', 'Version control', ' ...
377	Data and Analytics Associate	Business Intelligence and Analytics - Other,	Saint Gobain	4	1,238 Reviews	0 years	Chennai, Tamil Nadu	1	18 days ago	6,303	['Excel', 'SQL', 'Python', 'Data Analysis', 'D ...
1,500 rows x 11 columns											

FIG2 A dataset image.

Here 77 pages belong to A, and 1,500 pages, which is the total number of rows in our dataset, belong to B. When we open one card from page A, page B will open.

So, the total retrieval time for 1,500 rows is 54 min (approximately).

Data preprocessing

The preprocessing of the dataset is as follows:

- **Rating column:** For the “Rating” column, missing values were replaced with the average rating of the available data. This step ensures that all entries have a valid rating, maintaining consistency across the dataset.
- **Dropping the “Reviews” column:** The “Reviews” column was dropped from the dataset to streamline the analysis as it was not deemed essential for the scope of the project.
- **Experience column:** To standardize the “Experience” column, the minimum required experience was extracted, and missing values were replaced with “0” to represent entry-level positions.
- **Location standardization:** The “Locations” column was standardized by reducing variations. For example, locations like “Bangalore/Bengaluru” were unified to “Bangalore” for consistency and easier analysis.
- **Opening column:** Missing values in the “Openings” column were

replaced with “1,” assuming at least one opening exists for any posted job requirement.

- **Applicants column:** Entries of “Less than 10” applicants were standardized to “9” to maintain numerical consistency. All values in this column were converted to integers.
- **Skills column:** The “Skills” column was processed by converting all entries to lowercase and organizing them into lists of skills, making it easier to analyze and categorize skill requirements effectively. The preprocessed data are illustrated in Fig. 3.

Recommendation system overview

Purpose of the job recommendation system

The job recommendation system aims to bridge the gap between job seekers and suitable job opportunities by providing personalized recommendations tailored to individual candidate profiles.

Importance of matching candidates with suitable opportunities

Matching candidates with suitable job opportunities enhances job satisfaction, promotes career growth, and contributes to organizational success by aligning skills and preferences with job requirements.

Utilization of TF-IDF vectorization and cosine similarity

- **TF-IDF vectorization:** TF-IDF converts textual data into numerical vectors. It captures the importance of words in documents based on their frequency and rarity, enabling effective analysis and matching. This technique converts job descriptions and user-provided skills into numerical vectors. It captures the importance of words in the context of the entire dataset by considering both their frequency in a specific document (job description) and their rarity across all documents. This allows the system to prioritize more relevant terms that are indicative of the job’s requirements, thereby improving the representation of both job listings and user profiles.
- **Cosine similarity calculation:** Cosine similarity measures the similarity between two nonzero vectors in an inner product space. It quantifies the orientation similarity among vectors, facilitating the identification of relevant job positions based on user input. Once the job descriptions and user skills are represented as vectors, cosine similarity is used to measure the similarity among these vectors. This metric calculates the cosine of the angle between two vectors, providing a value between -1 and 1, where 1 indicates perfect similarity. By using

	Title	Job Role	Company	Rating	Experience	Locations	Openings	Applicants	Skill
0	Data Science Analytics Sr Analyst - Data Science	Data Science and Analytics - Other,	Accenture	4	5	Navi Mumbai	1	44	{research, data analysis, artificial intelligence ...
1	Data Science Professional	Data Science and Analytics - Other,	Davinta Financial Services	4.5	2	Bangalore/Bengaluru	1	387	{lgl, logistic regression, machine learning, d ...
2	Data Science-Application Developer	Data Science and Analytics - Other,	Accenture	4	5	Bengaluru	1	9	{tableau, computer operating, data analysis, m ...
3	Senior Analyst-Data Science	Data Science and Analytics - Other,	Resy	2.5	0	Gurgaon/Gurugram	1	1,065	{underwriting, financial services, risk manage ...
4	Consultant, Data Science and Analytics	Business Intelligence and Analytics - Other,	Transunion	4	5	Pune	1	16	{digital marketing, financial services, operat ...
...	...	...	...	...	...	...	...	...	...
372	Data Scientist	Data Scientist,	Benori Knowledge	3	3	Gurugram, Haryana	1	1,040	{artificial intelligence, machine learning, pr ...
374	Hiring For Data Scientist	Data Scientist,	Vidushi Infotech	3.2	4	Qatar	1	303	{r, neural networks, python, deep learning}
375	Data Modeller	Data Science and Analytics - Other,	Vishanz Business Services Private Limited	2.5	9	Hyderabad, Pune	10	9	{data modeling, erwin}
376	Data Scientist	Data Scientist,	Actionx Solutions	3.1	5	Pune	1	354	{data analysis, machine learning, database des ...
377	Data and Analytics Associate	Business Intelligence and Analytics - Other,	Saint Gobain	4	0	Chennai, Tamil Nadu	1	6,303	{data analysis, data visualization, sql, excel ...

1,389 rows x 9 columns

FIG3 Preprocessed data.

cosine similarity, the system can effectively identify how closely a user's skills align with the requirements of various job postings.

Together, these techniques enable the recommendation system to provide personalized job suggestions that are closely matched to the user's qualifications and preferences, thereby increasing the likelihood of successful job placements and enhancing user satisfaction.

Filtering by experience

The process of filtering by experience excludes job positions requiring more experience than the specified level. Cosine similarity scores are adjusted to ensure relevance.

Sorting and selection process

The sorting and selection process arranges job positions in descending order of cosine similarity to prioritize the most relevant matches. The ratio of job openings to applicants is calculated to determine the likelihood of selection:

$$\text{Selection probability} = \frac{\text{No. of openings}}{(\text{No. of applicants} + 1)}$$

Exploratory data analysis

As we have collected the latest data, we will perform exploratory data

analysis to gain insights and understand key trends in the dataset.

Distribution of skills across job roles

The sunburst chart in Fig. 4 visualizes the distribution of skills

across job roles in a hierarchical and radial format. This analysis offers valuable insights into the skill requirements and associations within our dataset, aiding in understanding the skill landscape for various positions.

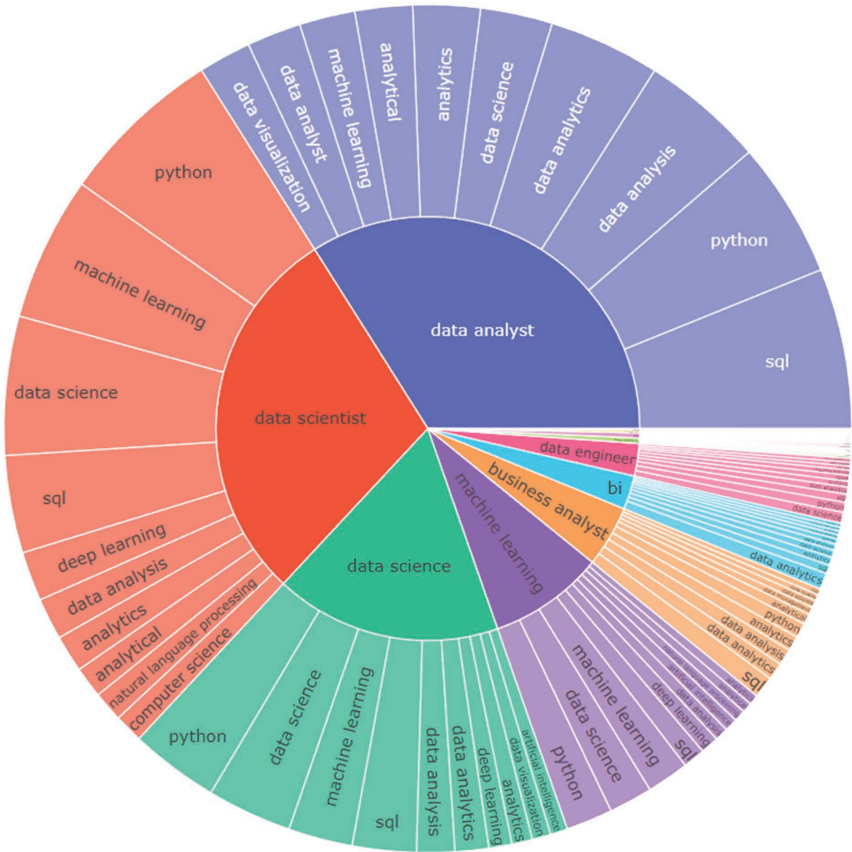


FIG4 The distribution of skills across job roles.

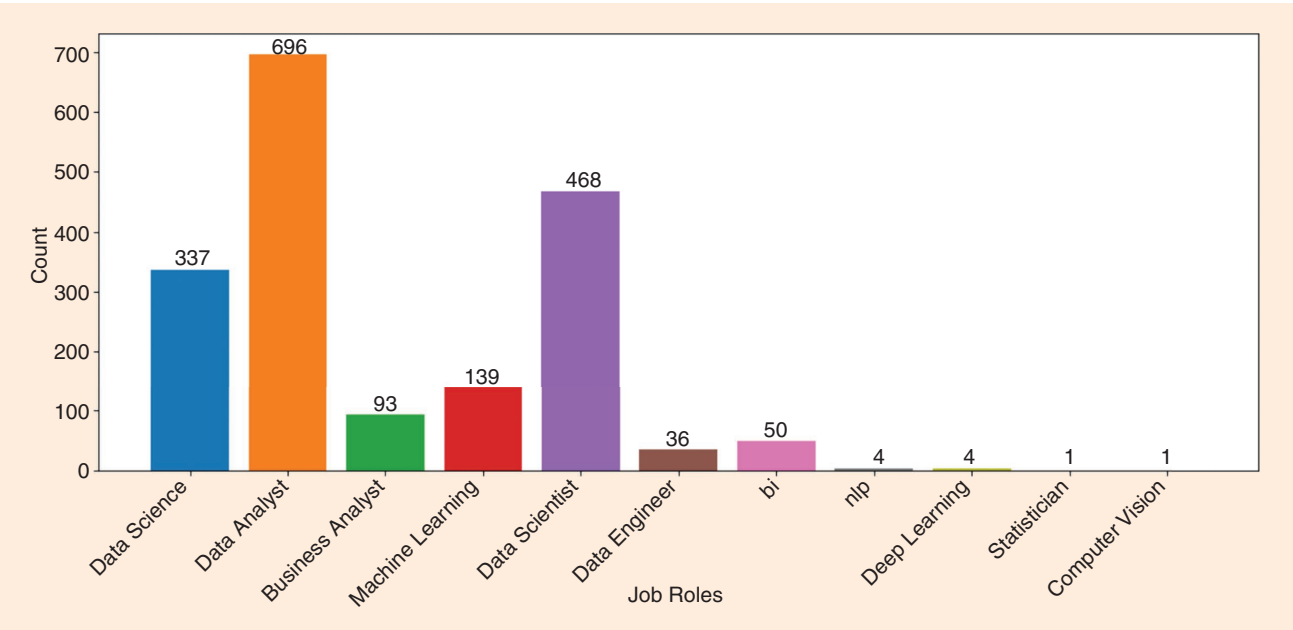


FIG5 The distribution of job roles.

Distribution of job roles

The bar graph in Fig. 5 showcasing the distribution of job roles illustrates the relationship between different job roles and their corresponding count, representing the number of available positions for each role.

Top skills

The pie chart in Fig. 6 displays the distribution of top skills based on their prevalence in job listings. Each slice represents a specific

skill, and the size of each slice corresponds to the proportion of job listings requiring that skill.

Top 10 job locations

This visualization in Fig. 7 provides a concise overview of the most prominent job locations based on the number of job openings. Each bar represents a specific location, and the length of the bar indicates the abundance of job opportunities in that area.

Openings versus applicants

The logarithmic bar graph in Fig. 8 compares the number of applicants with job openings, effectively visualizing the disparity between high applicant numbers and low job openings.

Conclusion

This research article presents a comprehensive endeavor to decode the nuances of the evolving job market in India through data-driven methodologies and advanced machine learning techniques.

Keys

Web scraping plays a pivotal role in enabling the comprehensive analysis and development of sophisticated recommendation systems within this article. By harnessing web scraping techniques, the project efficiently gathers real-time data from prominent job portals like Naukri.com, ensuring the compilation of a rich dataset reflective of the current job market trends in India. This dataset serves as the cornerstone for in-depth data analysis, facilitating the identification of emerging job roles, in-demand skills, geographic job distribution, and other critical insights. Through meticulous examination of job market dynamics, individuals and organizations gain valuable insights essential for informed decision making, career planning, and strategic workforce management. Furthermore, leveraging NLP techniques such as TF-IDF vectorization and cosine similarity, the project develops a sophisticated job recommendation system. This system utilizes the extracted data to match user-provided skills with job descriptions, offering personalized job recommendations tailored to individual preferences and expertise. Additionally, the system estimates the selection probability for candidates, providing valuable insights into potential career opportunities.

Future research

While the current iteration of the article yields significant value, there exist opportunities for future

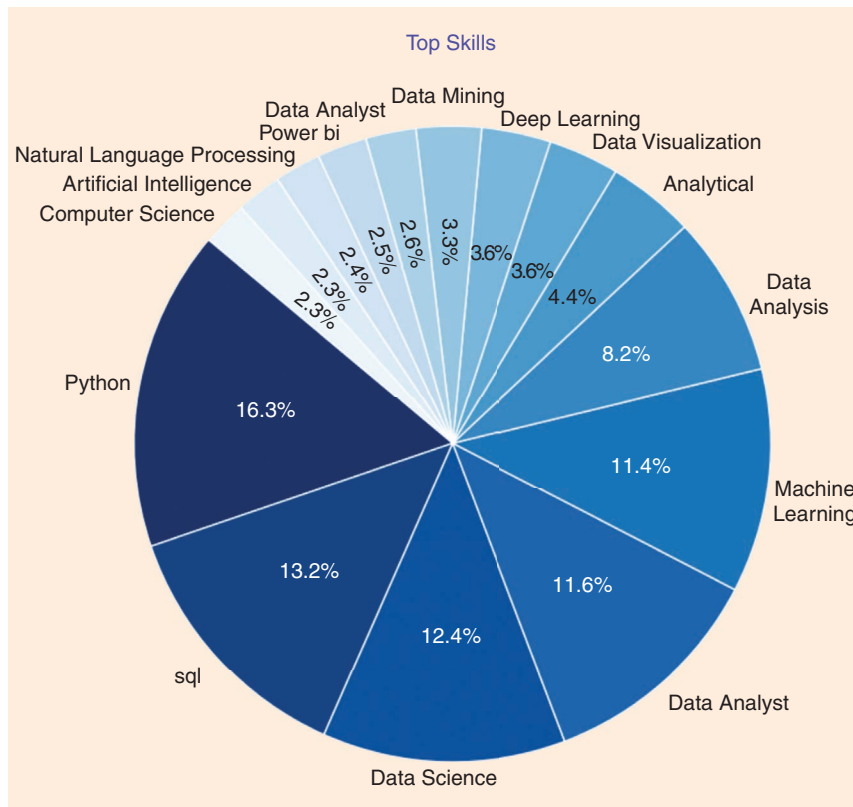


FIG6 The top skills.

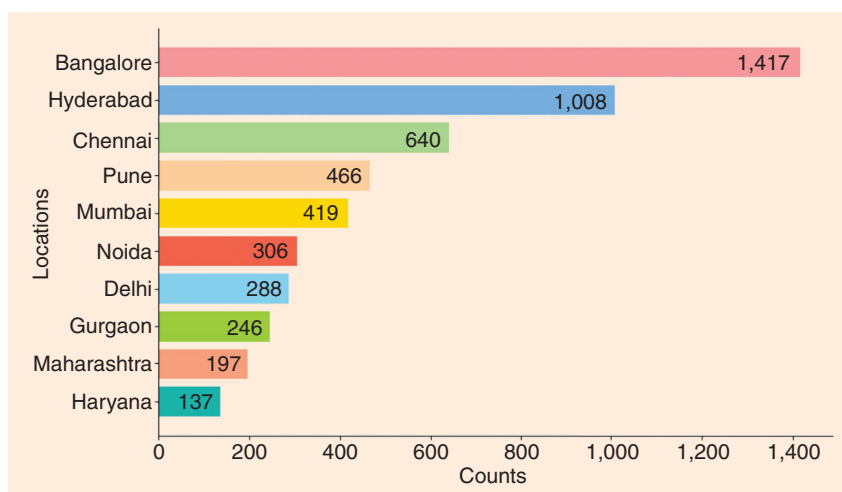


FIG7 The top 10 job locations.

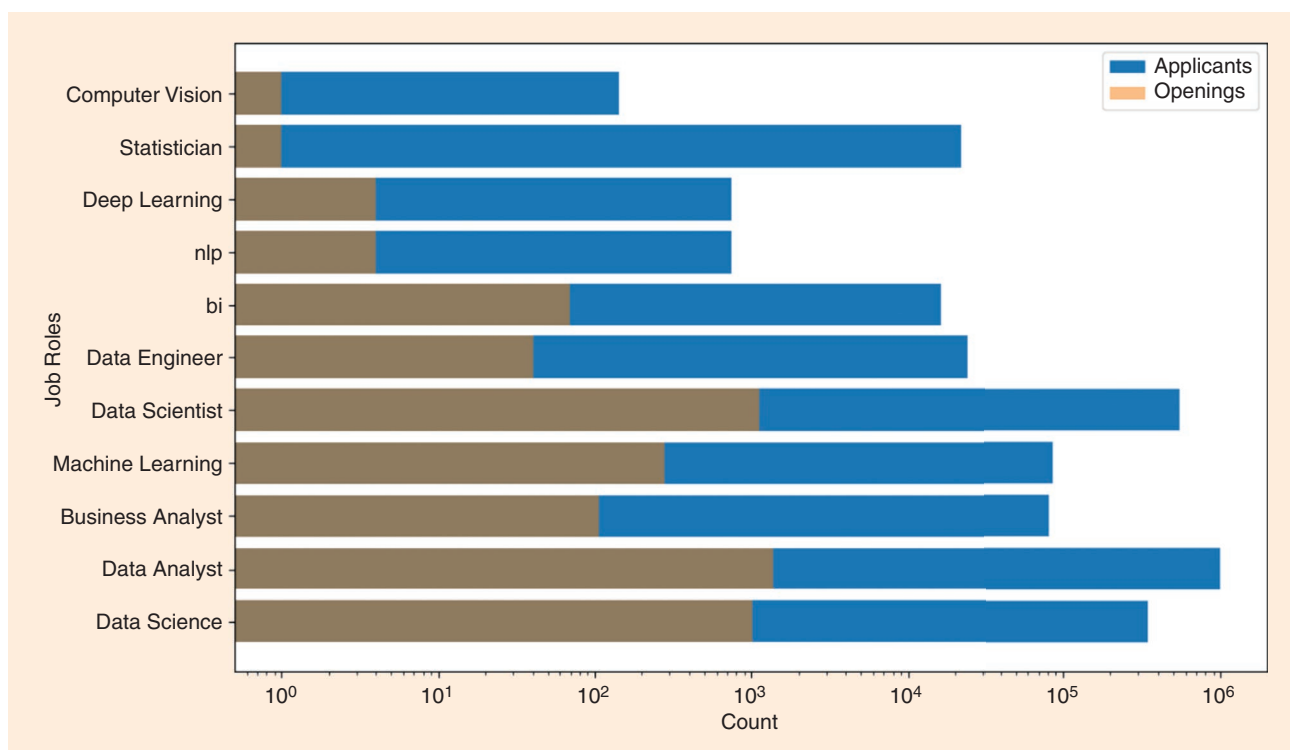


FIG8 Openings versus applicants.

improvements and enhancements. One potential avenue for improvement involves the integration of advanced machine learning algorithms, such as deep learning models, to further enhance the accuracy and effectiveness of the recommendation system. Furthermore, expanding the scope of data sources beyond different job portals to include social media platforms, industry reports, and government databases could provide a more comprehensive understanding of the job market landscape.

In conclusion, the article exemplifies the transformative potential of web scraping in enabling data-driven analysis and recommendation systems in the realm of job market dynamics. By continuously evolving and embracing innovations, the project stands poised to make significant contributions toward empowering individuals, organizations, and policy makers in navigating the complexities of the evolving job market landscape in India and beyond.

Acknowledgment

Saroja S is the corresponding author.

Read more about it

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Spotlight on the IEEE Region 8 Student Paper Contest

by Vera Markovic 

The IEEE Region 8 (R8) Student Paper Conference (SPC) is a prestigious competition for technical papers authored by IEEE Student and Graduate Student Members from student branches across Europe, Africa, and the Middle East.

The contest has a long-standing tradition of excellence; it was initiated by the second IEEE R8 Director, Jean Lebel, in 1967, just four years after IEEE R8 was established. Since then, it has been held annually without exception, even during the COVID-19 pandemic when fully online or hybrid formats were used.

The IEEE R8 SPC offers an excellent opportunity for undergraduate and master's students to engage in scientific research and technical writing. By working on an IEEE-related engineering topic and preparing a technical paper, students gain research experience that enhances their professional and personal development beyond traditional lectures.

In addition to student benefits, the contest also helps raise awareness of IEEE student branches; promotes IEEE engagement among students; and fosters collaboration within the academic community, including student branch counselors, supervisors, mentors, and reviewers.

The best way to ensure high-quality papers for the R8 contest is to organize local student paper contests at the IEEE student branch or section level. Each student branch may submit one paper per 100 members, with a maximum of three papers per student branch. Since 2022, the IEEE R8 Student Activities Committee has funded IEEE R8 sections and student branches to establish local student paper contests through the Local Student Paper Contest Events Program. This initiative



FIG1 The R8 SPC 2024 finalists, Matea Marinova, Loay Alfeqy, Jan Popič, Tim Bary, and Matic Pokornj, accompanied by Vera Markovic, SPC coordinator.

increases awareness and participation in the regional SPC by promoting local student paper contests.

Additionally, IEEE section officers can also encourage Student Members from branches without a local contest to submit their research directly to the regional SPC. Similarly, conferences hosting student paper sessions can serve as a source of SPC submissions, provided the paper is not published in the conference proceedings. The success of the SPC relies primarily on the engagement and enthusiasm of the student branch chairs and counselors, as well as section chairs and SA officers.

The deadline for paper submissions to IEEE R8 SPC is usually 1 December. An international jury of four experts from different IEEE R8 countries reviews the submitted papers through a blind evaluation process, which means that judges do not know the identity of the authors/institutions. After rigorous grading, the top five papers are selected for the Oral Finals, traditionally held at one of IEEE R8 flagship conferences, such as MELECON or EUROCON.

The SPC Oral Finals provide an invaluable opportunity for outstanding students to gain experience taking part in a prestigious IEEE conference. The same international jury evaluates the finalists, considering both the written and oral components to determine the final ranking and winners. Monetary prizes of US\$800, US\$500, and US\$200 are awarded to the first, second, and third place winners, respectively. In addition, the student branch of the first place winner receives the US\$250 Dick Poortvliet Award,



FIG2 The SPC finalists accompanied by jury members, 2024 IEEE President-Elect Kathleen Kramer, R8 vice chairs, and the SPC coordinator.

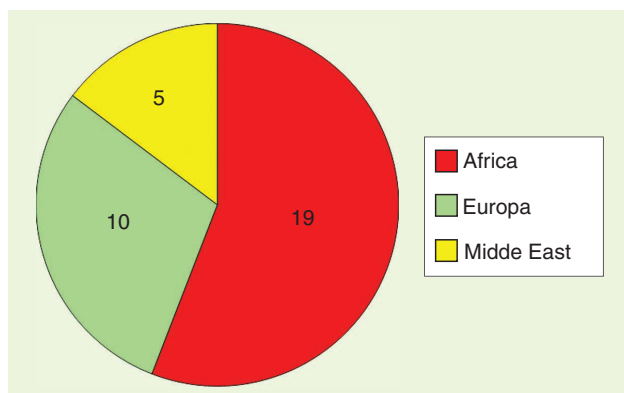


FIG3 Distribution of submitted papers from three subregions of IEEE R8.

honoring the 1995 Student Paper Contest coordinator who passed away during his term. As a special recognition, the papers of all five finalists are published in the conference proceedings and included in the IEEE *Xplore* digital library.

In 2024, the first SPC phase concluded with the selection of five best papers, authored by Matea Marinova from North Macedonia; Loay Alfeqy from Egypt; Tim Bary from Belgium; and Jan Popič and Matic Pokorn, both from Slovenia (Fig. 1).

The Oral Final was held during the IEEE R8 flagship conference, MELECON 2024. The city of Porto in Portugal served as the perfect venue, and conference organizers graciously hosted the finalists and jury members. Given the exceptional quality of the finalists' papers, selecting the winners was a challenging task for the SPC jury, which was composed of four professors: Paul Micallef from Malta, Gianfranco Chicco from Italy, Joao Carlos Ferreira from Portugal, and Vera Markovic from Serbia (Fig. 2).

The awards ceremony was held during the conference gala dinner aboard a cruise on the Douro River. It was a wonderful opportunity to acknowledge the outstanding achievements of our exceptional finalists. During the awards ceremony, it was a special honor to have the 2024 IEEE president-elect, Kathleen Kramer, on stage. She presented the First Prize in SPC 2024, which went to Tim Bary, Université Catholique de Louvain, Belgium, for his paper "Designing Pretraining Datasets From Unlabeled Data for EEG Classification with Transformers."

SPC 2025 is the 58th edition of the contest. We received approximately 40 submissions, of which 34 were deemed valid. Papers came from across R8, as shown in Fig. 3.

The written phase concluded in February 2025, with the selection of the top five papers advancing to the final round. This year, Prof. Krzysztof Gorecki from Gdynia Maritime University, Poland, joins the SPC jury, succeeding Prof. Paul Micallef, a long-time jury member and former SPC coordinator.

Listed below are the SPC 2025 finalists (in no particular order).

- Mohamed Achref Ben Ammar, Institut National des Sciences Appliquées et de Technologie-INSAT, Tunisia, "A Computational Approach to Modeling Conversational Systems: Analyzing Large-Scale Quasi-Patterned Dialogue Flows"
- Arman Nikraftar Khiabani, Catholic University of Louvain, Belgium, "Deep Reinforcement Learning for the Expert-Advice Multi-Armed Bandit"
- Emilija Chona, Ss. Cyril and Methodius University in Skopje, North Macedonia, "Transformer-Driven Dynamic Resource Reconfiguration in Cellular Networks"
- Klemen Berkovič, University of Maribor, Slovenia, "Cooperative Coevolution Framework with Recursive Differential Grouping Algorithms for Solving Large-Scale Optimization Problems"
- Zeyang Ai, Imperial College London, UK, "Interpretable Learning via Proximal Optimization."

The SPC 2025 Oral Finals will be held alongside the R8 flagship conference, EUROCON 2025, in Gdynia, Poland. As always, this event will be an exciting and enriching experience for the finalists, encouraging them to pursue further scientific research. We also hope that their success will inspire more IEEE Student Members to participate in this prestigious contest.

About the author

Vera Markovic (veram@ieee.org), retired, was a full professor with the Faculty of Electronic Engineering, University of Nis, 18000 Nis, Serbia. She is currently serving her fifth year as the IEEE R8 SPS coordinator and Student Activities Committee member. **P**

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IEEE Potentials is the IEEE's global magazine for students, with a circulation of 140,000. As technological advancements continue to reshape travel, Intelligent Transportation Systems (ITS) is pivotal in enhancing mobility, improving safety, reducing environmental impact, and fostering smarter, more sustainable transportation networks across all modes of transport. ITS integrates cutting-edge technologies, including artificial intelligence, IoT, big data, and communication networks, to optimize transportation systems ranging from railways and airways to maritime routes and urban transit.

On the ground, ITS revolutionizes road transportation through autonomous vehicles, smart traffic management, and connected infrastructure. In rail systems, it enables predictive maintenance, automated train control, and real-time passenger information systems. For aviation, ITS enhances air traffic management, airport operations, and flight safety systems. In maritime transportation, it facilitates vessel tracking, port automation, and route optimization. Urban mobility benefits from integrated solutions connecting buses, metros, trams, and micromobility options like shared bikes and scooters.

ITS has the potential to address critical global challenges such as urban congestion, transportation safety, greenhouse gas emissions, and accessibility across all transportation modes. Whether it's optimizing shipping routes to reduce maritime emissions, implementing smart railway signaling systems, or coordinating multimodal urban transit options, ITS is redefining how people and goods move worldwide.

As ITS gains momentum globally, discussing innovative approaches, sharing research insights, and exploring real-world applications across all transportation sectors is vital. The integration of different transport modes through intelligent systems creates seamless, efficient, and sustainable mobility networks. This special issue will allow experts, researchers, practitioners, and thought leaders to share their perspectives and expertise on this transformative field, encompassing innovations in land, air, and sea transportation systems. We are delighted to announce a call for articles for our upcoming magazine issue focusing on Intelligent Transportation Systems.

Topics of interest include, but are not limited to:

- Advanced ITS Simulation Systems
- Advanced Traffic & Travel Management
- Advanced Mobility Systems
- AI, ML and Deep learning for ITS
- Air, Road, and Rail Traffic Systems
- Behavioral Modeling in ITS
- Cooperative & Connected Vehicles
- Cyber-Physical-Social Systems for ITS
- Human Factors and HMI for ITS
- Human-Centered AI in ITS
- Intelligent Logistics
- Datasets for ITS
- Intelligent Waterborne Systems
- Internet of Things in ITS
- ITS Field Tests and Implementation
- ITS Validation and Verification
- Naturalistic Driving Data Analytics
- New Trends & Emerging ITS Technologies
- Public Policy, Regulatory & Societal Issues in ITS
- Safety, Security & Privacy in ITS
- Shared Mobility, Smart Mobility & Transportation 5.0

Submission of Papers: Prospective authors are invited to submit articles of up to 4,000 words. Articles should be aimed at a student audience and should not represent original research. The IEEE Potentials style requires no abstract, no citations from within the main text, and minimal equations but liberal use of relevant, captioned illustrations and photographs is encouraged. For more information, refer to the magazine's information for authors section. Articles for consideration should be submitted through <https://ieee.atyponrex.com/journal/pot-ieee>

Important Dates

Manuscript submission	1 July 2025
Notification of paper acceptance	1 August 2025
Accepted manuscript upload	1 September 2025
Publication date	November 2025

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Call for Papers

Theme Issue on “Cybersecurity, AI Ethics, and Tech Governance”

IEEE Potentials Theme Issue
September/October 2025

Lead Editors:

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IEEE Potential is the IEEE's global magazine for students, with a circulation of over 140,000 readers. Our upcoming themed issue will focus on *Cybersecurity, AI Ethics, and Tech Governance*. We invite submissions that focus on these critical areas, encouraging authors to explore innovative ideas and solutions. We especially encourage IEEE volunteers at all levels to share their outreach efforts to raise awareness of important AI ethics and cybersecurity issues.

Topics of interest include, but are not limited to:

- Emerging trends in cybersecurity
- Ethical considerations in AI
- Privacy concerns in the era of big data
- Bias mitigation strategies in machine learning algorithms
- Responsible development of autonomous systems and robotics
- Regulatory frameworks on cybersecurity and AI ethics
- Cybersecurity policies for critical infrastructure protection
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- Social impacts of AI
- Human-computer interaction

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